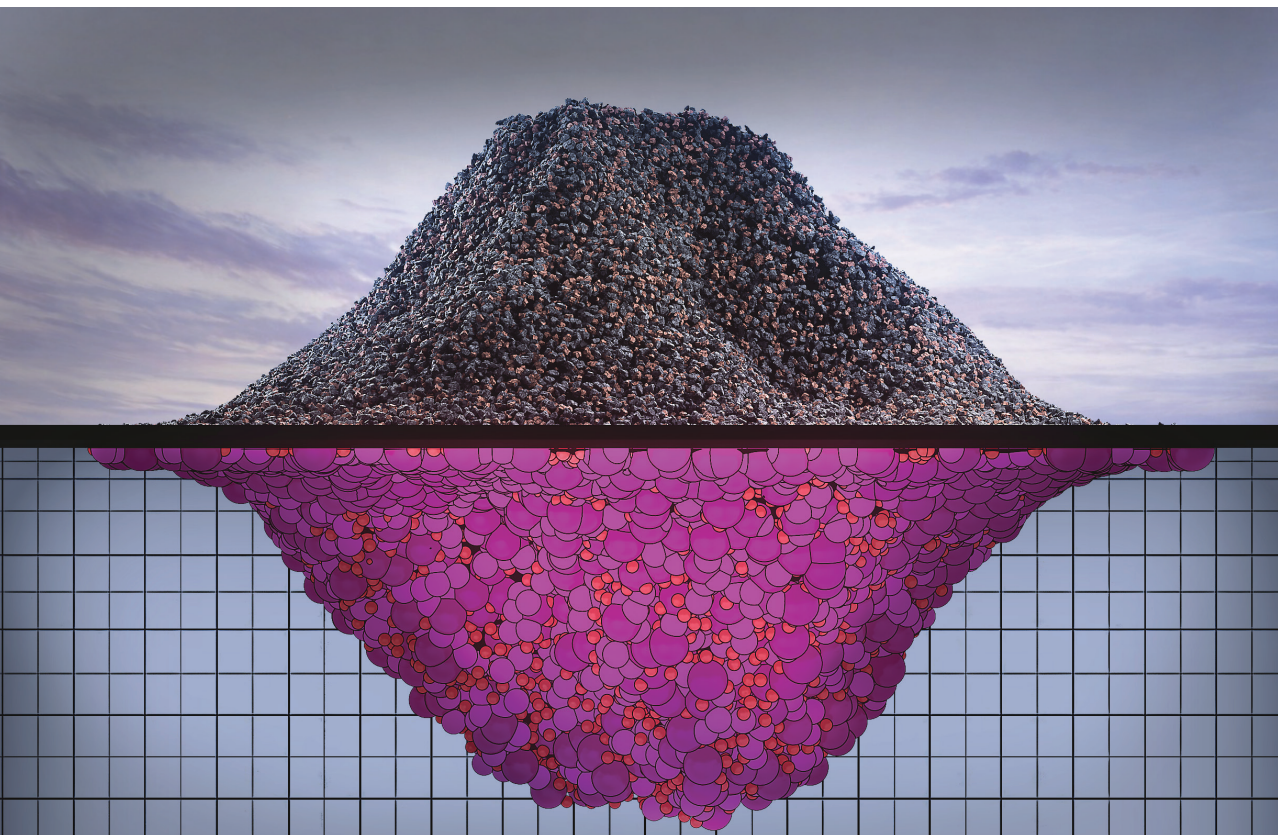
Doctoral Thesis in Civil and Architectural Engineering

Aggregate Fracture in Unbound Road Materials

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Abstract

Mineral aggregate fracture in unbound granular materials (UGMs) beneath asphalt pavement layers affect pavement performance and may accelerate distresses. Improving the understanding, characterization, and quantitative prediction of aggregate fracture in UGMs is therefore essential, particularly to support the use of marginal-quality aggregates in road construction and thereby reduce the demand for high-quality aggregates and lower the environmental impact of road infrastructure.

This thesis introduces a new numerical framework, combined with an experimental study, to predict aggregate fracture in UGMs and quantify its influence on UGM macro-mechanical behavior. The framework is based on the discrete element method (DEM) and enables evaluation of aggregate fracture for varying gradations, loading conditions, and aggregate types. To ensure general applicability, granular mechanics–based contact laws and statistical fracture models are developed and incorporated into DEM.

The model parameters are identified and the framework validated through laterally confined monotonic uniaxial compression tests on UGMs. The tested materials included different aggregate types and gradations and were subjected to different maximum compressive loads. For UGMs composed of crushed granite, the DEM model captures the effects of gradation and load magnitude on both macro-mechanical response and aggregate fracture. To extend the framework to a wider range of aggregates, particularly marginal-quality aggregates, a new particle fracture model is developed that accounts for aggregate shape variability and statistical volume effects on fracture force distributions. The model is evaluated using single-particle crushing tests on four aggregate types and compared with two widely used fracture models, showing improved agreement with measured aggregate strength. When implemented in the DEM framework, the new model improves fracture predictions for UGMs containing marginal-quality aggregates.

The feasibility of using DEM to assess how aggregate fracture affects elastic stiffness and permanent deformation resistance of UGM is evaluated. Emphasis is put on UGMs containing marginal aggregates and on the potential for optimizing pavement structural design to enable their use

without excessive performance loss. Blended UGMs containing crushed granite and crushed brick are investigated using confined compression tests and X-ray CT, and the observations are incorporated into the DEM model to predict both macro-mechanical behavior and aggregate fracture.

A systematic analysis of aggregate fracture in UGMs subjected to confined monotonic compression tests is conducted to identify the governing factors of fracture. The results show that aggregate fracture is controlled by the coupled effects of aggregate strength, gradation, and applied load, with the strength and applied load identified as the dominant factors. The results further demonstrate that aggregate fracture in UGMs cannot be fully predicted using standard aggregate strength tests alone and should therefore be evaluated under field-representative gradations and loading conditions. To support this assessment, an approach combining the developed DEM models with experimental measurements is introduced, and a fracture evolution parameter is proposed to quantify the progression of aggregate fracture during compression. The parameter shows strong agreement with experimentally observed aggregate fracture and provides an effective means for characterizing fracture development in UGMs under loading.

In summary, the results demonstrate that the developed DEM framework can quantify how a wide range of aggregate types—including marginal-quality aggregates—along with gradation and loading, affect UGM performance and aggregate fracture, and can support performance-based material selection and pavement design to mitigate aggregate crushing.

Keywords

Discrete Element Method (DEM); unbound granular material (UGM); aggregates; marginal aggregates; aggregate fracture; X-Ray computed tomography.

Sammanfattning

Brott i aggregat i obundna granulära material (OGM) under asfaltbeläggningslager påverkar vägbeläggningens prestanda och kan påskynda skador. Förbättrad förståelse, karakterisering, och kvantitativ prediktering av aggregatbrott i GGM är därför avgörande, särskilt för att möjliggöra användning av material med lägre kvalitet i vägbyggnad. Detta kan minska behovet av högkvalitativa och nybrutna ballastmaterial och därmed sänka väginfrastrukturens miljöpåverkan.

Denna avhandling introducerar ett nytt numeriskt ramverk, kombinerat med en experimentell studie, för att förutsäga aggregatbrott i OGM och kvantifiera dess inverkan på materialets makromekaniska beteende. Ramverket bygger på diskreta elementmetoden (DEM) och möjliggör utvärdering av aggregatbrott för varierande graderingar, belastningsförhållanden, och aggregattyper. För att säkerställa att ramverket är generellt tillämpbart har kontaktlagar baserade på kontaktmekanik samt statistiska brottmodeller utvecklats och integreras i DEM.

Modellparametrar identifieras och modellerna valideras genom inneslutna uniaxiella kompressionstester på OGM bestående av olika aggregattyper och graderingar vilka utsätts olika maximala tryckbelastningar. För OGM bestående av krossad granit fångar DEM-modellen effekterna av kornkurva och belastningsnivå på både makromekaniskt beteende och aggregatbrott. För att utvidga ramverket till ett bredare spektrum av aggregat, särskilt material av lägre kvalitet, utvecklas en ny partikelbrottsmodell som tar hänsyn till variationer i aggregatens form och statistiska volymeffekter på brottkraftsfördelningar. Modellen utvärderas med krossningstester på enskilda partiklar av fyra olika aggregattyper och jämförs med två vanligt använda brottsmodeller, vilket visar förbättrad överensstämmelse med uppmätta brottkrafter. När modellen implementeras i DEM-ramverket förbättras prediktionen av brott i OGM som innehåller aggregat av lägre kvalitet.

Möjligheten att använda DEM för att bedöma hur aggregatbrott påverkar elastisk styvhet och motstånd mot permanent deformation utvärderas, med fokus på OGM som innehåller aggregat av lägre kvalitet och potentialen att

optimera vägens konstruktionsutformning för att möjliggöra deras användning utan betydande prestandaförlust. Blandade OGM bestående av krossad granit och krossat tegel undersöks vidare genom inneslutna kompressionstester och röntgen-CT, och observationerna integreras i DEM-modellen för att förutsäga både makromekaniskt beteende och aggregatbrott.

En systematisk analys av aggregatbrott i OGM utsatta för inneslutna monotona kompressionstester genomförs för att identifiera de styrande faktorerna bakom brott. Resultaten visar att aggregatbrott styrs av de samverkande effekterna av aggregatstyrka, kornkurva och applicerad last, där Los Angeles-värdet (LA) och maximal last identifieras som de dominerande faktorerna. Resultaten visar vidare att aggregatbrott i OGM inte kan predikteras enbart med standardiserade index och därför bör utvärderas under fältrepresentativa kornkurvor och belastningsförhållanden. För att stödja denna bedömning introduceras en metod som kombinerar de utvecklade DEM-modellerna med experimentella mätningar, och en parameter för skadesutveckling föreslås för att kvantifiera progressionen av aggregatbrott under kompression. Parametern visar stark överensstämmelse med experimentellt observerat aggregatbrott och erbjuder ett effektivt sätt att karakterisera brottsutveckling i UGM under belastning.

Sammanfattningsvis visar resultaten att det utvecklade DEM-ramverket kan kvantifiera hur aggregattyp, kornkurva och belastning påverkar OGM's prestanda och aggregatbrott, och kan stödja prestandabaserat materialval och vägutformning för att minska krossning av aggregat i obundet granulärt material.

Nyckelord

Diskret elementmetod (DEM); obundet granulärt material (OGM); aggregat; aggregat av lägre kvalitet; aggregatbrott; röntgentomografi (X-ray CT).

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Stockholm, February 2026
M. Kaan Etikan

List of Publications

This thesis is based upon the work presented in the following peer-reviewed journal articles:

Paper I

Etikan, M. K., Jelagin, D., Olsson, E. & Partl, M. N. (2024a). Experimental and numerical analyses of crushing resistance of unbound road materials. *International Journal of Pavement Engineering*, 25(1), 2330630. https://doi.org/10.1080/10298436.2024.2330630

Paper II

Etikan, M. K., Jelagin, D., Partl, M. N., & Olsson, E. (2025a). Discrete element analysis on aggregate breakage of unbound granular road materials. Canadian Journal of Civil Engineering, 52(7), 1379-1390. https://doi.org/10.1139/cjce-2024-0091

Paper III

Etikan, M. K., Jelagin, D., Olsson, E., & Partl, M. N. (2025b). A new statistical fracture model for particles in unbound road materials. Materials and Structures, 58(6), 228. https://doi.org/10.1617/s11527-025-02760-4

Paper IV

Etikan, M. K., Jelagin, D., Olsson, E., Partl, M. N., Tan Z. (2026a). Modelling and Testing of Marginal Aggregate Fracture in Unbound Granular Materials. Manuscript is submitted for publication to Construction and Building Materials.

Paper V

Etikan, M. K., Jelagin, D., Partl, M. N. Olsson, E. (2026b). Fracture Testing of Aggregates in Unbound Granular Road Materials. Manuscript is submitted for publication to Road Materials and Pavement Design.

All appended papers were written jointly by the first author and the co-authors.

In Papers I–III and V, the first author carried out the experimental work, developed the numerical model, performed the numerical simulations, and, together with the co-authors, conducted the data analysis.

In Paper IV, the image analysis of the computed tomography (CT) measurements was performed by Dr. Zhifei Tan. The first author conducted the mechanical tests, performed the numerical simulations, and, together with the co-authors, evaluated the experimental and modelling results.

In addition to the appended papers, the following conference proceeding publications have been produced as part of the project:

- Etikan, M. K., Jelagin, D., Olsson, E., & Partl, M. N. (2023a). Modelling and Testing of Aggregate Crushing in Unbound Road Materials. *Advances in Materials and Pavement Performance Prediction (AM3P 2022)*.

- Etikan, M. K., Jelagin, D., Olsson, E., & Partl, M. N. (2023b). Discrete Element and Experimental Analysis of Marginal Road Materials. *VIII International Conference on Particle-Based Methods (PARTICLES 2023)*.

- Etikan, M.K., Jelagin, D., Partl, M.N., Olsson, E. (2024b). Discrete Element Analysis on Aggregate Breakage of Marginal Unbound Road Materials. *In: Carter, A., Vasconcelos, K., Dave, E. (eds) 14th International Conference on Asphalt Pavements ISAP2024 Montreal. ISAP APE 2024*. Springer, Cham. https://doi.org/10.1007/978-3-031-67252-1_119

- Etikan, M. K., Jelagin, D., Olsson, E., & Partl, M. N. (2025c). A Numerical Study of the Effect of Aggregate Fracture on the Unbound Granular Materials Response under Triaxial Loading. In: Eberhardsteiner, L., Hofko, B. and Blab, R. (eds.) *Advances in Materials and Pavement Performance Prediction IV: Contributions to the 4th International*

Conference on AM3P 2025. Vienna: TU Wien Academic Press, pp. 492–495. https://doi.org/10.34726/10780

- Etikan, M. K., Jelagin, D., Olsson, E., & Partl, M. N. (2026c). Discrete Element Analysis of the Response of Unbound Granular Materials Consisting Marginal Road Materials under Triaxial Loading. *In Proceedings of the Twelfth International Conference on the Bearing Capacity of Roads, Railways, and Airfields BCRRA 2026*. Ljubljana.

- Etikan, M.K., Jelagin, D., Partl, M.N., Olsson, E. (2026d). A New Methodology for Fracture Testing of Unbound Granular Road Materials. *In Proceedings of the 2nd RILEM International Symposium on Bituminous Materials ISBM 2026*. Padova.

Contents

1. Introduction 1
1.1. Aim, Objectives and Limitations 5
1.2. Outline of the Thesis 9
2. Methodology 11
2.1. Computational Methodology 11
2.1.1. Model of confined monotonic uniaxial compression test on UGM 13
2.1.2. Contact laws 15
2.1.3. Aggregate fracture models 19
2.2. Experimental Methodology 24
2.2.1. Aggregate Types and Gradations 24
2.2.2. Confined uniaxial compression test 26
2.2.3. Single particle crushing test 28
2.2.4. X-Ray computed tomography measurements and image analysis 30
2.2.5. New parameter to characterize aggregate fracture from compressive test measurements 32
3. Results and Discussion 35
3.1. Confined Compression Tests on UGMs 36
3.1.1. UGMs containing standard aggregates 36
3.1.2. UGMs containing marginal strength aggregates 39
3.1.3. X-ray CT image analysis on blended UGMs 44
3.1.4. Effects of UGM material parameters on aggregate fracture in UGMs 47
3.2. DEM Modelling of Confined Compression Tests 50
3.3. DEM Modelling of Aggregate Fracture Under Cyclic Loading 55
3.4. Single Particle Fracture Tests and Extended Fracture Model 58
3.5. DEM Modelling of UGMs Containing Marginal-Quality Aggregates 61
3.6. Determination of Aggregate Fracture Evolution Parameter 68
4. Conclusions and Future Work 73
5. References 77

1. Introduction

Unbound granular materials (UGMs) are widely used in the granular layers of roads, and their mechanical behaviour strongly influences deterioration rates and overall functional performance. Road construction worldwide consumes very large volumes of UGMs, making optimization essential to reduce both economic costs and environmental impacts of road infrastructure. Such optimization requires a thorough understanding of how key UGM parameters—such as gradation, aggregate type, and in-service load levels—relate to UGM performance within the road structure.

One important aspect affecting UGM performance—particularly with respect to permanent deformation accumulation, elastic stiffness evolution, and drainage capacity—is the propensity of UGM aggregates to fracture under external loading. Aggregate fracture in UGMs within the unbound layers of asphalt road structures can lead to distresses such as rutting, fatigue cracking, and surface unevenness (Colagrande et al., 2011; Mahmoud & Masad, 2007; Zeghal, 2009). These issues reduce pavement service life and functionality, posing economic and social challenges through high maintenance costs and public safety risks (Moretti et al., 2012; Ragnoli et al., 2018). Therefore, a comprehensive understanding of aggregate fracture performance in UGMs is essential for optimizing material selection, structural design, and performance predictions.

As the availability of high-quality material resources declines, the use of aggregates of marginal-quality becomes increasingly necessary (Partl, 2019). In this context, understanding aggregate fracture in UGMs is particularly important for promoting the use of marginal-quality aggregates in road construction. This approach can reduce the demand for high-quality virgin aggregates and lower the environmental impact and carbon footprint of road construction (Barbhuiya et al., 2025; de Andrade Salgado & de Andrade Silva, 2022). However, marginal-quality aggregates often exhibit lower resistance to fracture (Erlingsson et al., 2017); therefore, rigorous evaluation is required

for identifying both suitable UGM compositions and the maximum load levels within the pavement structure at which these aggregates can be applied without compromising structural performance (Wang et al., 2024; Zhang et al., 2021).

Several empirical methods assess aggregate fracture resistance in the laboratory, including the Micro-Deval, Los Angeles abrasion, and aggregate crushing value tests (Bjarnason et al., 1999; Saeed et al., 2001). These tests provide practical indices for ranking aggregates based on fracture resistance, establishing minimum acceptability criteria. However, their correlation with in-field aggregate performance is weak (Wu et al., 1998).

Furthermore, empirical tests do not fully capture realistic loading conditions experienced by aggregates in UGMs. In UGMs, external forces are transmitted through a network of aggregate contacts, which is determined by aggregate size distribution in the material along with their size and shape. Thus, in addition to geometric and mechanical aggregate properties, fracture susceptibility is influenced by the magnitude and distribution of contact forces at stone-to-stone contact points. The conventional procedures typically evaluate fracture resistance on a selected fraction size of the aggregates, most commonly the coarse fraction. Although this fraction is often the most susceptible to fracture, restricting the evaluation to a limited section of the gradation curve neglects the role of the overall particle size distribution on the fracture behaviour, cf. Bjarnason et al. (1999). Thus, empirical tests may not sufficiently account for the effects of loading conditions or UGM parameters such as gradation and aggregate strength on crushing resistance. Furthermore, those tests may not be suitable for evaluating aggregate fracture in blended UGMs composed of several aggregate types (e.g., blends of good aggregates with marginal-quality ones). These limitations highlight the need for more advanced approaches to accurately measure and predict aggregate behaviour in UGMs.

The Discrete Element Method (DEM), developed by Cundall and Strack (1979), is a powerful tool for quantifying the relationship between applied loads, aggregate mechanical properties, and gradation in UGMs. It allows for analysing aggregate deformation and damage induced by stone-to-stone contacts while linking micromechanical damage at contact points to the macroscopic behaviour of UGMs (Cervera et al., 2017; Cole & Peters, 2008).

However, application of DEM has computational limitations regarding size, number and shape of particles, since it is difficult to account for gradations with high amount of fines due to calculation time and also fully capture different shapes of aggregates. Therefore, it is still necessary to work with idealizing and simplifying assumptions.

Both 2D and 3D DEM studies have emerged as valuable tools for investigating aggregate fracture in pavements (Cui & Bhattacharya, 2017; Vallejo et al., 2006) and railway ballast (Lu & McDowell, 2010). Vallejo et al. (2006) used DEM to examine where aggregate fracture initiates and how it develops within the unbound granular layer of asphalt pavements. Cui and Bhattacharya (2017) explored the impact of aggregate fracture on pavement permeability. Lu and McDowell (2010) employed 3D DEM to study how interlocking and asperity fracture in railway ballast affect UGM performance under triaxial loading. These studies highlight DEM's usefulness in capturing both the macroscopic effects of aggregate fracture on UGMs and the micromechanical mechanisms governing particle breakage. DEM thus offers a method to establish a quantitative link between the macro-mechanical behaviour of UGMs and the micromechanical processes occurring at stone-to-stone contact points.

Recently, DEM has also been applied to study the aggregate fracture in UGMs composed of recycled construction and demolition materials (Afshar et al., 2017; Oskooei et al., 2021b; Oskooei et al., 2021a; Zhang et al., 2021). These materials are increasingly used in UGMs to reduce the environmental impact of road construction (de Andrade Salgado & de Andrade Silva, 2022; Kong et al., 2023). However, they often exhibit lower strength and inferior shape characteristics compared to conventional road materials, making them more susceptible to fracture. Optimizing material selection and structural design is therefore crucial to integrate these materials without significantly compromising performance. The influence of aggregate shape on fracture behaviour has been evaluated using 3D DEM for waste rock (Afshar et al., 2017), recycled concrete (Oskooei et al., 2021a), and crushed brick aggregate (Oskooei et al., 2021b). Furthermore, Zhang et al. (2021) investigated the feasibility of using DEM simulations to optimize UGM designs incorporating different recycled materials, such as crushed brick, gravel, and mortar. Their

findings suggest that DEM can be a valuable tool for optimizing blend ratios of recycled and virgin aggregates to maintain adequate UGM performance.

Several studies have also examined the effect of different particle fracture criteria in DEM simulations on computational predictions. de Bono and McDowell (2016) evaluated various fracture criteria for silica sand, finding that the choice of fracture criterion significantly influences computational accuracy. Their results showed that the maximum contact force and octahedral shear stress failure criteria provided the best correlation with experimental data. Bisht and Das (2021) used 2D DEM modelling to investigate how aggregate shape evolution after fracture affects UGM macroscopic behaviour. They compared two particle fracture modelling approaches: the agglomerate method and the particle replacement method, which substitutes large particles with smaller ones upon fracture. Their findings indicated that both axial stress-strain response and void ratio in UGMs were significantly affected by aggregate fracture and the choice of fracture modelling technique.

While previous research on DEM modelling of aggregate fracture in UGMs has yielded promising findings, several gaps remain. In particular, current DEM studies on aggregate fracture primarily rely on contact and fracture models calibrated through experimental observations of UGMs at the macro-scale. While these models provide insight into the mechanisms and material parameters governing aggregate fracture performance, their reliance on empirically calibrated contact and fracture laws limits the extrapolation of results to a broader range of conditions, including different aggregate types, loading magnitudes, and structural designs of unbound road layers. This limitation is particularly problematic when assessing the incorporation of marginal-quality aggregates into blended UGMs composed of both high- and marginal-quality constituents. In practice, blending can enable the use of marginal-quality materials while maintaining acceptable performance; however, the fracture response of these aggregates—both individually and within blended systems—remains insufficiently understood from both experimental and modelling perspectives.

Additionally, a comprehensive analysis of the relation between aggregate fracture and the macro-mechanical behaviour of UGMs is still lacking in the literature. These limitations hinder the application of DEM modelling for

analysing and optimizing new UGMs and road structural designs. A particularly important aspect, in terms of optimizing the use of natural resources, is leveraging DEM modelling to refine material selection and composition and structural designs to incorporate marginal-quality aggregates into UGMs without significantly compromising performance.

1.1. Aim, Objectives and Limitations

The aim of this thesis is to improve the understanding, characterization, and quantitative prediction of aggregate fracture in unbound granular materials (UGMs) of standard and marginal quality for roads. This is pursued by developing a numerical framework for predicting aggregate fracture and quantifying its impact on the macro-mechanical behaviour of UGMs. The framework is intended to provide reliable predictions across a broad range of loading conditions and material configurations, accounting for variations in gradation and aggregate type. This requires physically grounded particle contact and fracture laws that capture the mechanics of contact-induced deformation and damage at the aggregate level.

To assess the framework's potential for optimizing UGMs containing marginal-quality aggregates, aggregate fracture is investigated experimentally in blends of standard and marginal-quality aggregates. The framework's ability to capture the effects of UGM composition and load magnitude on aggregate fracture is then evaluated. Experimental validation is a central component of the work and is carried out using uniaxial monotonic confined compression tests on UGM specimens together with crushing tests on individual aggregates. X-ray computed tomography (CT) was used to gain further insight into aggregate fracture in one blended UGM by characterizing the locations of fractured aggregates and the sizes of the resulting fragments within the material. In addition, to mitigate limitations of conventional aggregate fracture tests discussed in previous section, a new fracture evolution parameter is introduced, derived from the difference between experimental measurements with fracture and the fictitious "no-fracture" response simulated with the developed DEM framework. The proposed parameter exhibits a unique relationship with experimentally measured aggregate breakage and can therefore be used to track the evolution of aggregate fracture in UGMs under applied loading.

Accordingly, the specific objectives of this thesis can be listed as follows:

1. Development of particle fracture laws for DEM simulations capturing the mechanics of aggregate fracture for a wide range of mineral aggregate types. An important aspect of this problem is also the development of an efficient methodology for determining the material parameters for the developed fracture laws based on the laboratory measurements.

2. Development and validation of a DEM modelling framework for predicting aggregate fracture in UGMs, while incorporating contact and fracture laws accurately describing the grain-scale interactions between the aggregates. In terms of validation, the attention is focused on the frameworks' capability of capturing both macro-scale material response as well as local phenomena such as aggregate fracture and aggregate re-arrangement.

3. Quantitative evaluation of the effect of aggregate fracture on UGMs macro-scale response under monotonic and cyclic axial loading. The attention is focused on the effects of loading conditions and of UGMs gradations and aggregate type, on the elastic and permanent deformation performance of UGMs.

4. Evaluation of aggregate fracture in blended UGMs containing both standard and marginal-quality aggregates, and assessment of the developed DEM framework's ability to predict fracture behaviour in such blended systems.

5. Experimental investigation of how UGM gradation, aggregate type, and applied load magnitude influence aggregate fracture. Development of a testing and analysis methodology capable of capturing aggregate fracture in UGMs under progressively increasing loading.

The workflow of this thesis is illustrated in Figure 1. Papers I and III address the first two objectives outlined above. In Paper I, a DEM modelling

approach is developed and implemented in the commercial DEM software PFC3D™ (Itasca Consulting Group, 2019) to study aggregate fracture in UGMs. This approach incorporates particle contact laws based on granular mechanics, as recently proposed by Olsson et al. (2019a; 2019b).

In Paper I, a particle failure model based on the Weibull weakest-link approach is employed to investigate the effects of UGM gradation and of loading magnitude on the aggregate crushing in UGMs composed of granite aggregates commonly used in unbound base layers in Sweden. Additionally, a simple procedure for identifying contact and failure law parameters from experimental data is proposed. Using these parameters, the model's ability to capture the effects of gradation and loading on aggregate breakage in UGMs is examined.

To extend the applicability of the DEM modelling framework to a broader range of aggregate types, Paper III introduces a new particle fracture model. The proposed fracture model combines Weibull weakest link approach with a distribution that accounts for the uncertainty caused by the large variety in aggregate shapes. The developed model is validated based on single particle crushing tests as well as with results of confined uniaxial compression tests performed on UGMs composed of four different aggregate types.

To address the third objective, Paper II employs the developed DEM model to examine how aggregate fracture influences the macro-scale response of UGM, focusing on elastic and permanent deformations under monotonic and cyclic compressive loading. Feasibility of using the developed DEM approach to optimize the placement of UGMs in unbound road layers is evaluated.

Paper IV addresses the fourth objective by investigating aggregate fracture in blended UGMs, in which one size fraction of a conventional road aggregate is replaced with marginal-quality aggregates. Fracture behaviour in blended UGMs is evaluated experimentally and compared with that of UGMs composed entirely of conventional and entirely marginal-quality aggregates. In addition, Paper IV evaluates the ability of the DEM framework to capture the effects of UGM composition and load magnitude on fracture, and demonstrates how the model can be used to explore blend proportions under expected loading conditions.

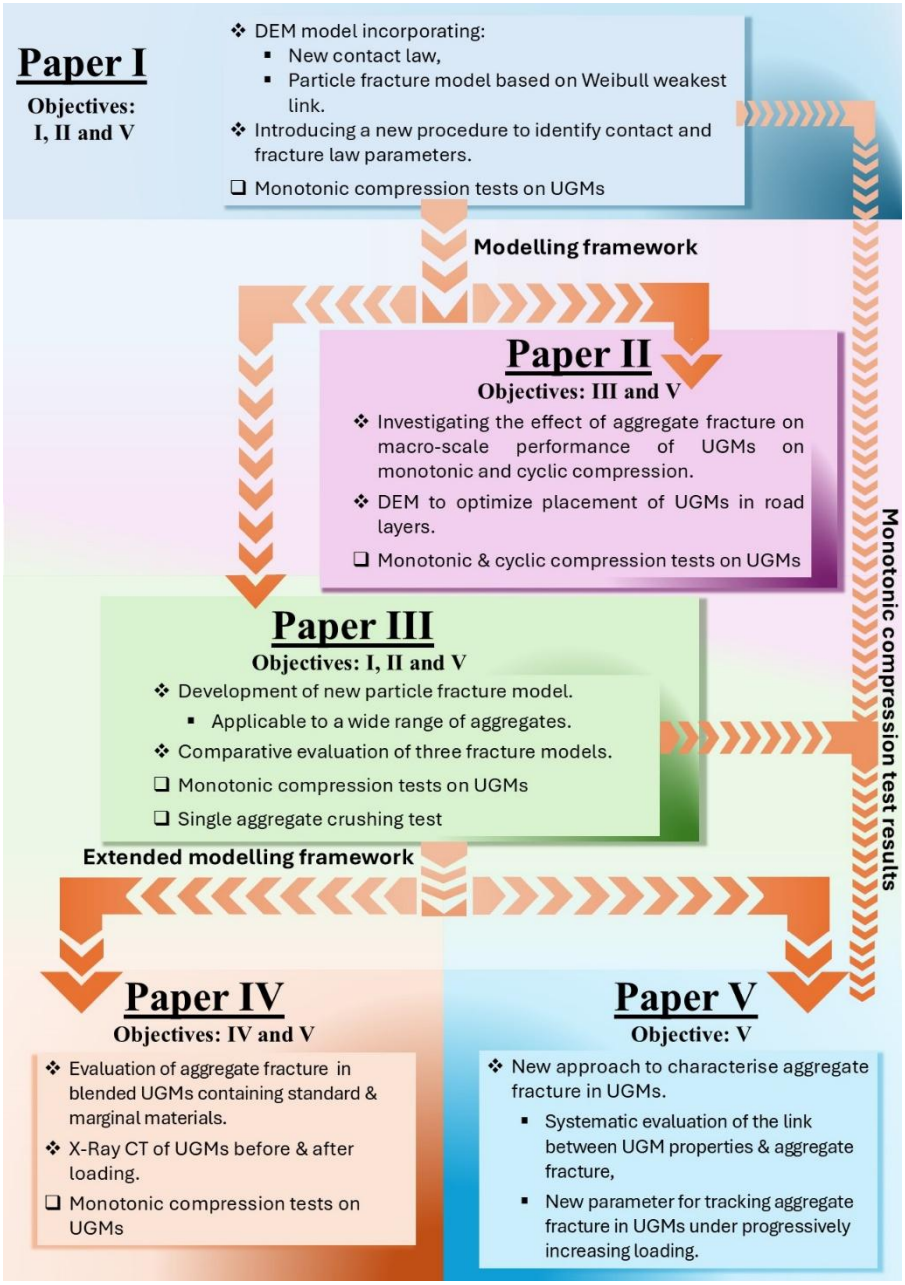


Figure 1. Diagram of the workflow of the research presented in this thesis.

The fifth objective is addressed across Papers I–V through confined uniaxial compression tests performed on UGMs of different compositions

and at multiple load levels. In Papers I–IV, aggregate fracture is quantified by measuring changes in UGM gradation before and after testing. X-ray computed tomography (CT) is further used in Paper IV to examine fracture patterns, the spatial distribution of fractured aggregates, and fragment size distributions. In Paper V, the experimental results from Papers I and III are used to systematically assess how UGM properties—gradation, aggregate type, and Los Angeles (LA) abrasion value—affect fracture under varying load levels. Finally, Paper V proposes and evaluates a new fracture evolution parameter suitable for tracking aggregate fracture in UGMs subjected to progressively increasing loading.

The research presented in this thesis is subjected to several limitations. First of all, the developed modelling approach relies on spherical particles to represent aggregates. While computationally efficient, this simplification precludes the possibility of fully evaluating the effects of aggregate shape on the mechanical behaviour of UGMs. In addition, fully representing road UGM gradations would require inclusion of fine particles, which may significantly alter aggregate interactions—particularly after fracture. Moreover, field aggregates experience varying moisture conditions that influence UGM mechanical behaviour; these aspects will be addressed in future work.

1.2. Outline of the Thesis

This thesis begins with an introductory chapter, where the objectives, background and method of investigation are presented. In Chapter 2, the computational and experimental methodologies are described. Chapter 3 presents and discusses the results obtained from measurements and simulations. Finally, in Chapter 4 the main conclusions are presented along with suggestions for future work.

2. Methodology

This chapter presents the methodology used in this thesis and is divided into two main Sections: computational methodology and experimental methodology, described in Sections 2.1 and 2.2, respectively.

In the computational methodology Section 2.1, first the basic principles of the discrete element method (DEM) are introduced. Subsection 2.1.1 contains a summary of the applied modelling approach, followed by Subsections 2.1.2 and 2.1.3 where the contact laws and failure models for describing the contact-induced deformation and damage of the aggregates are presented in detail.

In the experimental methodology Section 2.2, the aggregate types and UGM gradations used to investigate aggregate fracture are summarized in Subsection 2.2.1. Aggregate fracture is investigated experimentally at two different scales. To investigate the aggregate fracture within UGMs, confined uniaxial compression tests were conducted as described in Subsection 2.2.2. Furthermore, to evaluate accuracy of different fracture models presented in Subsection 2.1.3 and for identifying their parameters, single particle crushing tests were performed on aggregate particles of different sizes. They are presented in Subsection 2.2.3. Fracture behaviour within UGMs is further examined using X-ray CT, as presented in Section 2.2.4. Finally, Subsection 2.2.5 introduces an alternative methodology developed in this thesis to quantify aggregate fracture in UGMs under progressively increasing loading.

Throughout this thesis, the term “aggregates” will refer to the stones used in the experiments, while the term “particles” will be used to describe the stone surrogates employed in the DEM simulations.

2.1. Computational Methodology

The DEM, originally developed by Cundall and Strack (1979), is used to investigate granular materials, where individual particles interact through

contact points. In this method, particles are modelled as individual objects, and their positions and velocities are determined through explicit numerical integration of Newton's second law of motion. The forces acting on the particles include gravity and the contact forces resulting from interactions with other particles or surrounding wall elements defining the specimen boundaries. A particle is considered to be mechanically interacting when it overlaps with another particle or a wall, and the magnitude of overlap determines the magnitude of the contact force.

In the context of DEM modelling of UGMs, the basic approach is summarized in Figure 2. As illustrated, UGMs are characterized by their material parameters, such as gradation, porosity, and aggregate properties, e.g. stiffness and toughness. Those material parameters are represented in the DEM as a collection of particles, of a certain shape, following the same size distribution as the aggregates in the UGM. In this thesis, spherical particles are used to represent aggregates to minimize the number of modelling parameters and ensure reasonable computational efficiency. It has to be pointed out, however, that this assumption may be removed in a relatively straightforward manner, by using clumps of spherical particles to represent arbitrary shapes. The mechanical properties of the aggregates influence the UGM response in the DEM model by affecting the contact law and fracture model parameters. Once established, the DEM model can be used to evaluate the impact of UGM material parameters on macro-scale responses, such as elastic stiffness and permanent deformation resistance as well as on local phenomena like aggregate fracture.

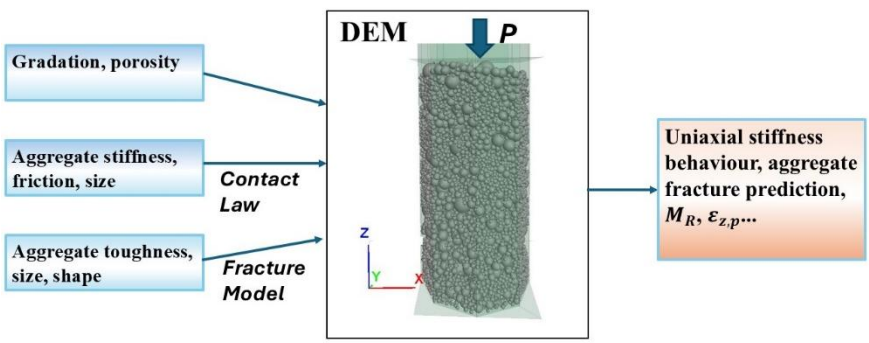


Figure 2. General workflow of DEM modelling approach.

2.1.1. Model of confined monotonic uniaxial compression test on UGM

This thesis mainly deals with DEM modelling of aggregate fracture in UGMs subjected to confined monotonic uniaxial compression tests, presented in detail in Chapter 3. Accordingly, the model geometry and loading conditions are defined following the test conditions. Examples of generated DEM models are shown in Figure 3 along with the boundary and loading conditions as well as the coordinate system used. The model is developed using a commercial DEM software PFC3D™. User subroutines for granular mechanics-based contact law and fracture models are incorporated into this commercial software using C++ and FISH coding, respectively.

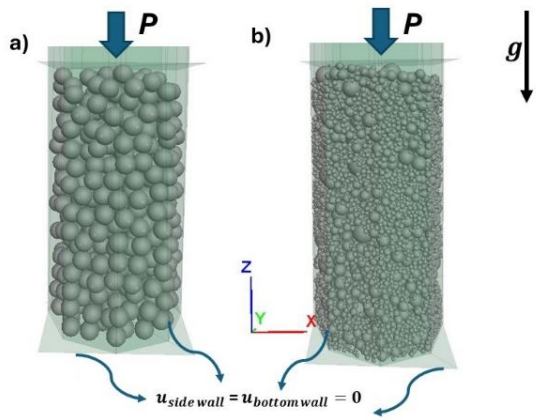


Figure 3. DEM models of UGMs with gradations ranging from a) 11.2 - 16 mm, b) 2 - 16 mm.

The modelling flowchart is presented in Figure 4. The main modelling steps are as follows:

1. First, a confining cylindrical mould is generated using wall elements. The confinement follows the dimension of the mould used in experiments except its height is chosen significantly higher, so that the particles can be generated without overlapping each other. The displacement of the wall elements, u, are set to zero. Then, the particles are generated at random positions within the mould. Each particle is characterized by a specific Young's modulus, E, density, ρ, friction coefficient, μ, and local damping coefficient, η. To model the interlock effect between particles, the rotational degrees of freedom of the particles are constrained.

2. Gravity g is applied to allow particles to fall to the bottom of the cylinder as displayed in Figure 4. To ensure that particles do not acquire unrealistically high accelerations, local damping coefficient, η, as implemented by PFC3D™ is introduced. Damping allows the rapid attainment of steady state condition. As the modelling is performed under quasi-static conditions, the introduced damping does not affect the calculations significantly, as discussed in Paper I. At this stage, porosity of the DEM model is controlled to match the porosity of the UGM.

3. After the particles are settled at the bottom of the mould, the UGM is compressed in the vertical direction using a rigid lateral wall element until a pre-defined maximum applied load, P_{max}, is reached. The simulated UGM is compressed incrementally, with the normal and the tangential contact forces for each particle in contact being calculated at each increment according to the contact laws introduced in Subsection 2.1.2.

4. In order to determine particle fracture, contact forces arising at each contact point are evaluated at each load increment, ΔP, cf. Figure 4, using the fracture models introduced in Subsection 2.1.3 as the UGM is compressed. If the contact forces exceed the particle fracture strength, which is calculated statistically using the aggregate fracture models introduced in Subsection 2.1.3, the particle is registered as fractured (purple spheres in Figure 4), and its mechanical response is modified following the procedure presented in Subsection 2.1.3.

5. The effect of aggregate fracture on the macroscopic response of UGMs is analysed along with the amount of the aggregate fractures accumulated and the results are compared to experiments.

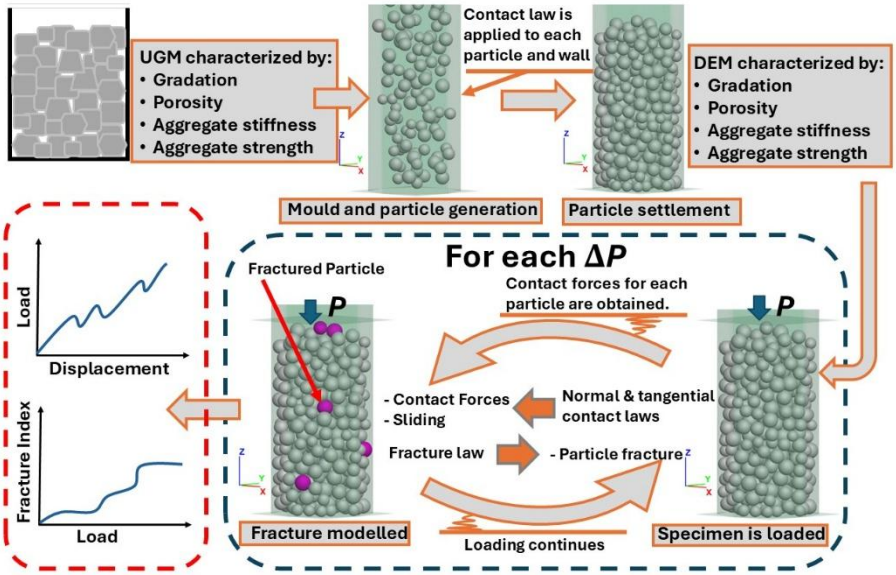


Figure 4. Modelling flowchart.

As this study primarily focuses on capturing the phenomena of aggregate fracture, the accurate resolution of contact forces between the aggregates is of outmost importance. Accordingly, the computational procedure presented in Figure 4, incorporates contact laws based on granular mechanics which are presented in Subsection 2.1.2. The fracture models used to determine and model the aggregate fracture are presented in Subsection 2.1.3 below.

2.1.2. Contact laws

The contact forces governing particle interaction can be separated into a normal contact force, F_N, and a tangential contact force, F_t, as shown in Figure 5. The magnitudes of these forces are governed by the overlap between contacting particles, denoted by h, and the tangential displacement between them, denoted by δ. In order to resolve the equilibrium in DEM simulations, normal and tangential contact laws, linking the F_N and h, and F_t and δ respectively are required. The normal contact law used in this thesis is based on the work by Olsson et al. (2019a), where a contact law capturing accurately aggregate deformations at the contact points has been developed based on the granite spherical indention tests on a flat surface conducted by Cervera et al. (2017). The results of indentation tests, where a spherical indenter pressed

against the flat surface of granite materials, may be adapted to the case of two particles in contact in DEM, by introducing the effective contact radius, R_0, as follows:

$$R_0 = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1}, \tag{1}$$

where R_1 and R_2 represents the radii of the two particles in contact, as shown in Figure 5.

The normal contact force described by Olsson et al. (2019a) consist of three stages that correspond to different physical processes observed in the experiments of Cervera et al. (2017), as presented in the Equation (2).

$$F_N = \begin{cases} \dfrac{F_s}{h_s} h, & h \le h_s \\ k_p\,(h - h_1)^{1.5}, & h_s \le h \le h_L, \\ k_p\,(h_L - h_1)^{1.5} + 1.5\,k_p\,\sqrt{h_L - h_1}\,(h - h_L), & h_L \le h \end{cases} \tag{2}$$

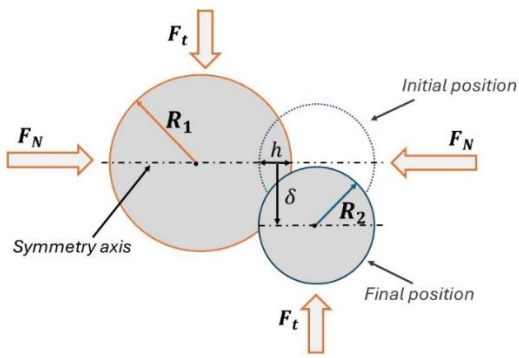


Figure 5. Representation of two spheres in contact and variables required to define contact - overlap relationship.

As shown in Figure 6, the contact process starts with crushing of surface asperities, which is represented by a linear relationship until the surface asperity depth, h_s, is reached. Then, F_N exhibits Hertzian elastic contact behaviour until initiation of quasi-plastic deformation at the depth h_L. Subsequently, F_N deviates from the Hertzian behaviour at higher forces, demonstrating linear force displacement relationship due to shear driven quasi-plastic damage. The values of h_s and h_L observed in the experiments of Cervera et al. (2017) were 0.02 mm and 0.08 mm, respectively.

The force observed at the indentation depth, h_s, is denoted as F_s, and in the indentation experiments by Cervera et al. (2017) for granite it was observed to be 100 N, with an indenter of 6.25 mm radius. Subsequently, F_s is scaled with $\sqrt{R_0}$ to ensure a continuous transition for particles of different size, as shown below:

$$F_s = 100\ \text{N} \sqrt{\frac{R_0}{6.25\ \text{mm}}}, \tag{3}$$

A continuous contact force through the different stages as a function of h is established by introducing h_1, defined by

$$h_1 = h_s - \left(\frac{F_s}{k_p}\right)^{2/3}, \tag{4}$$

where k_p represents the contact stiffness as described in Hertz contact theory (Hertz, 1881):

$$k_p = \frac{4E_{eff}\sqrt{R_0}}{3}, \tag{5}$$

and E_{eff} describes the effective Young's modulus of two particles in contact, as given by:

$$E_{eff} = \left(\frac{1-v_1^2}{E_1} + \frac{1-v_2^2}{E_2}\right)^{-1}, \tag{6}$$

where E_1 ,E_2, v_1, v_2 are Young's moduli and Poisson's ratios of particles in contact, respectively. In the simulations, h_l is adjusted using the particle radii to achieve a continuous transition of force, as described below:

$$h_L = 0.08\ \text{mm}\ \frac{R_0}{6.25\ \text{mm}} + h_s, \tag{7}$$

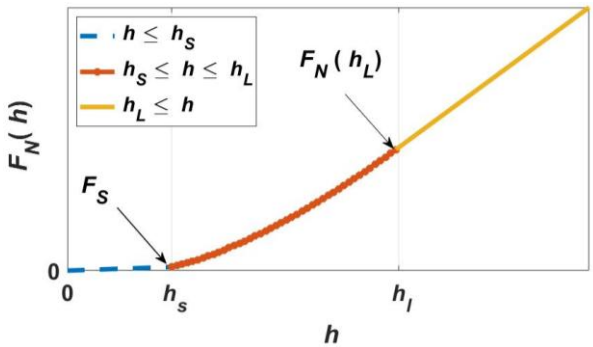


Figure 6. The normal force - overlap relationship (h_s denotes the surface asperity depth and h_l represents the depth where quasi plastic deformation initiates).

The resulting $F_N(h)$ is illustrated in Figure 6. The linear dashed blue line represents the asperity crushing region, the red line is denoting the elastic Hertzian behaviour, and the yellow line shows the quasi-plastic region, represented by the relationships in rows one to three of Eq. (2) respectively.

The tangential contact force, F_t, determines the frictional sliding behaviour of particles in DEM, with respect to the friction limit, μF_N, where μ represents the friction coefficient. The sliding behaviour depends on the relationship between F_t and μF_N, as shown in Eq. (8).

$$F_t = \begin{cases} k_s\delta, & \|F_t\| \leq \mu F_N \quad no\ slip \\ \mu F_N, & \|F_t\| > \mu F_N \quad slip \end{cases}, \tag{8}$$

where δ is the tangential displacement between particles in contact, and k_s denotes the tangential shear stiffness, as expressed by Eq. (9). Due to variations in tangential stiffness, spurious increases in elastic energy may occur during some contact displacement histories. To address this issue, tangential shear stiffness is scaled as proposed by Agnolin and Roux (2007), as shown as:

$$k_s = \frac{2(1-\nu_{eff})}{2-\nu_{eff}}\ 1.5\ k_p^{\frac{2}{3}}\ F_N^{\frac{1}{3}}, \tag{9}$$

where ν_{eff} denotes the effective Poisson's ratio. The effective Poisson's ratio of the contacting particles is calculated in PFC3D™ as to align with Hertz-Mindlin contact model (Mindlin & Deresiewicz, 1953):

$$v_{eff} = \frac{E_2(v_1^2 + v_1) + E_1(v_2^2 + v_2)}{E_2(v_1 + 1) + E_1(v_2 + 1)}, \quad (10)$$

As seen from as Equations (1) - (10) the contact behaviour of particles is determined by their E, μ, ν.

2.1.3. Aggregate fracture models

To accurately predict aggregate fracture in UGMs, the predicted contact force distributions must be combined with a statistical representation of aggregate strength. In DEM, aggregate fracture is simulated using a fracture model to evaluate the macro-mechanical behaviour of the UGM and gain micromechanical insights into its fracture mechanisms. To predict mechanical behaviour of various materials, aggregate types, and gradations under different loading conditions, it is essential to employ a statistical aggregate fracture model that describes the statistical distribution of aggregate fracture forces, F_f, based on aggregate properties.

In this thesis, three aggregate fracture models were evaluated with respect to their capability to capture aggregate fracture in UGMs under various conditions. Those three models are presented below along with the basic theoretical considerations behind them.

Fracture of an aggregate occurs when the stress induced by external loading exceeds the strength of the material somewhere within an aggregate. This local material strength is controlled by the presence of small local defects and inhomogeneities, such as voids, microcracks, and locally weak material. As a result, two primary sources of scatter in fracture force for aggregates can be identified. The first is the effect of aggregate shape, which can significantly influence the stress distribution, $\sigma(x, y, z)$, within the aggregate when subjected to the same external force. The second source is the effect of aggregate size, which influences the presence of the random material defects, in terms of their size, number and location. This introduces a size dependent variability of aggregate fracture strength, as larger aggregates are more likely to encounter defects of a specific size compared to smaller aggregates. Fracture models of varying complexity may be used to account for the scatter in aggregate strength with a different degree of accuracy.

The first fracture model, Model A, employs the classic Weibull distribution with the cumulative density distribution function S_A for the aggregate fracture force, F_f, as expressed in following form:

$$S_A = 1 - \exp\left[-\left(\frac{F_f}{\sigma_A R^2}\right)^{m_A}\right], \tag{11}$$

where σ_A denotes the Weibull stress that governs the median strength of the particles, and m_A is the Weibull modulus that controls the scatter of the distribution. This model disregards any statistical size dependency, as indicated by the use of the conveniently defined nominal fracture stress, $\sigma_f = F_f/R^2$, where $2R$ represents the mean sieve size of the aggregates. Consequently, it only considers scatter in fracture force resulting from shape effects.

An alternative method for expressing the probability of aggregate fracture force is the weakest link method (Weibull, 1951). This method (Model B) calculates the probability of fracture for a small volume and then integrates it over the entire volume. A Weibull distribution is utilized for the weakest link method, leading to the cumulative density function S_B for Model B, as follows:

$$S_B = 1 - \exp\left[-\int_V \left(\frac{\sigma(x,y,z)}{\sigma_{WB}}\right)^{m_B} \frac{dV}{V_{ref}}\right], \tag{12}$$

where σ_{WB} and m_B represent material parameters, and V_{ref} denotes an arbitrarily chosen reference volume introduced for dimensional consistency. The integration is conducted solely within regions where tensile stresses are present. In practice, the integration process is cumbersome, as a detailed information of the stress state is required, and a substantial number of different aggregates needs to be tested to ensure statistical validity. By employing the normalized variables $\sigma(x,y,z) = \tilde{\sigma}(x,y,z)\sigma_f = \tilde{\sigma}(x,y,z)F_f/R^2$ and $V = R^3\tilde{V}$ in Eq. (12), the expression for the classical weakest-link integral becomes:

$$S_B = 1 - \exp\left[-\left(\frac{\sigma_f}{\sigma_{WB}}\right)^{m_B} \frac{R^3}{V_{ref}} \int_{\tilde{V}} \left(\tilde{\sigma}(x,y,z)\right)^{m_B} d\tilde{V}\right], \tag{13}$$

The integral part is denoted by I^{m_B} that is expressed as:

$$I^{m_B} = \int_{\tilde{V}} \left(\tilde{\sigma}(x,y,z)\right)^{m_B} d\tilde{V}, \tag{14}$$

where the integral I may vary over multiple orders of magnitude. Assuming that the integral of I results in a constant value, the Weibull weakest-link distribution for Model B can be expressed as:

$$S_B = 1 - \exp\left[-\left(\frac{F_f}{\sigma_{WB}R^2}I\right)^{m_B}\frac{R^3}{V_{ref}}\right] = 1 - \exp\left[-\left(\frac{F_f}{\sigma_B R^2}\right)^{m_B}\frac{R^3}{V_{ref}}\right], \tag{15}$$

where σ_B represents the Weibull stress that is calculated as $\sigma_B = \sigma_{WB}/I$, and m_B is the Weibull modulus for Model B. Both material parameters are interpreted similarly to those in Model A**.** However, in Model B, the governing factor for the variability in fracture forces is assumed to be the size effect, which influences the randomness in the distribution of defects within the aggregates.

It should be noted that the actual shape of the aggregates introduces considerable variability into the integral, I^{m_B}, due to the wide variety of particle geometries. This uncertainty is addressed in the derivation of Model C. Given the limited information available regarding the values of this integral, $I^{m_B}$is considered a stochastic variable. For simplicity, a two-parameter Weibull distribution is proposed for its probability density function $p(I)$, as follows:

$$p(I) = \frac{nI^{n-1}}{I_0^n}\exp\left[-\left(\frac{I}{I_0}\right)^n\right], \tag{16}$$

where I_0 determines the median value of I and n regulates the scatter similar to Eq. (11). Additionally, the size of the aggregates remains to be uncertain. In Model A, the aggregate size of $2R$ was assumed to represent the mean size between the smaller and the larger sieves. However, a clear distribution exists between these sizes. This distribution is incorporated into Model C by employing a probability distribution for the particle radius, denoted as $p(R)$. The aggregate size between adjacent sieve sizes is assumed to be uniformly distributed in terms of their volume; thus, $p(R)$ is expressed as follows:

$$p(R) = \frac{2R_N^2 R_{N+1}^2}{(R_{N+1}^2 - R_N^2)R^3}, \tag{17}$$

where $2R_N$ and $2R_{N+1}$ represents the smaller and larger sieve size respectively. When the stochastic variables of the integral I and the particle size R are introduced in Eq. (15), along with the probability distribution from Eq. (16) and (17), the resulting distribution for the fracture force becomes:

$$S_C = \int_{R_N}^{R_{N+1}} \int_0^\infty p(R)p(I)\left(1 - \exp\left[-\left(\frac{F_f}{\sigma_{WC}R^2}I\right)^{m_C}\frac{R^3}{V_{ref}}\right]\right) dI dR, \tag{18}$$

where σ_{WC} and m_C are the material parameters. Detailed information about geometry and loading condition of aggregate is required to determine the value of I. Therefore, a normalized variable ξ is introduced, defined as $\xi = I/I_0$. This leads to a new distribution for ξ, expressed as:

$$p(\xi) = \frac{n_C}{I_0}\xi^{n_C-1}\exp[-\xi^{n_C}], \tag{19}$$

The value I becomes deterministic when $n_c \to \infty$. This results in the elimination of variability due to I in the classical weakest-link model, transforming Model B into the deterministic model given in the right-hand side of Eq. (15). Furthermore, σ_C is introduced as a material parameter through the relationship of $\sigma_C = \sigma_{WC}/I_0$, leading to the fracture force distribution referred to as Model C, expressed as follows:

$$S_C = 1 - \int_{R_1}^{R_2} \int_0^\infty p(\xi)p(R)\exp\left[-\left(\frac{F_f}{\sigma_C R^2}\xi\right)^{m_C}\frac{R^3}{V_{ref}}\right] I_0 d\xi\, dR. \tag{20}$$

In this distribution, the explicit dependency on the value I_0 is negated, as the I_0 terms next to $d\xi$ and in the denominator of $p(\xi)$ cancels each other out.

The fracture force distribution captured by Model C accounts for particle size effect addressed by the weakest-link model, through the probability distribution $p(R)$, and shape variations effect through $p(\xi)$. Additionally, this model introduces another material parameter, n_C, in comparison to the other two models. In summary, Model A incorporates the shape effect to capture the scatter of fracture forces of aggregates, with the parameters σ_A and m_A. Model B addresses the size effect to capture the scatter of fracture forces of aggregates using the parameters σ_B and m_B, while Model C simultaneously

accounts for both the size and the shape effects, using three material parameters σ_C, m_C and n.

In Papers I and II, the Weibull weakest link model, i.e. Model B above, is implemented in DEM through FISH coding, based on the work by Olsson et al. (2019b; 2020). The resulting DEM model is used to investigate the effect of gradation and loading magnitude on aggregate crushing performance of two types of granite aggregates. Furthermore, in Paper II, the effect of aggregate fracture on macro-scale response of UGMs is evaluated. In Paper III, the new statistical force model, Model C, is introduced to predict aggregate breakage in UGMs composed of aggregates with marginal strength characteristics. This new model is also systematically evaluated in comparison with Models A and B in Paper III.

It remains to discuss how the post-fracture behaviour of aggregates is modelled in DEM. Capturing this post-fracture behaviour is essential for obtaining a realistic response of the UGMs.

To address this issue, several methods are implemented in the literature. One widely used approach is the agglomerate method, where a particle is modelled as a collection of sub-particles bonded together (Cheng et al., 2003). This method enables a detailed representation of the post fracture behaviour of the particle but incurs large computational costs as the number of particles in the simulation is by magnitudes larger than the number of aggregates. Another approach is the particle replacement method, where a large particle is subdivided into smaller bonded or unbonded particles once it fractures. However, as noted by Guo et al. (2020), the particle replacement method presents challenges in mass and energy conservations.

Given the challenges of the two methods above, an alternative approach introduced by Olsson and Larsson (2015) is employed in this thesis, where the stiffness of a particle is reduced upon fracture:

$$E_{fractured} = C_w \cdot E_{initial}, \tag{21}$$

where C_w represents the weakening coefficient with a value ranging between 0 and 1. Reducing the initial Young's modulus, $E_{initial}$, results in a decrease in contact stiffness, this allows to simulate local particle rearrangement caused by fracture. According to Olsson and Larsson (2015), this method results in a

realistic response provided that only a small proportion of particles fractures, and no large movements occur. This is exactly the case for the problem at hand, as only a relatively small fraction of aggregates may be expected to fracture in UGMs under practically relevant loading conditions.

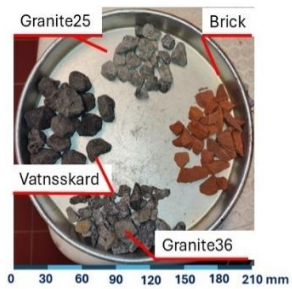
2.2. Experimental Methodology

2.2.1. Aggregate Types and Gradations

The materials used in this study include two crushed granite types, one with relatively high and the other with relatively low fracture resistances, crushed brick, and crushed volcanic aggregates. The high-strength granite aggregates were mined from Arlanda quarry in Sweden, while the weak granite aggregates were mined from a paragneiss in the southern part of the Västerbotten region in Sweden. The crushed brick aggregates, composed of crushed old brick roof tiles were obtained commercially. The volcanic aggregates were from Vatnsskard in Iceland. These volcanic aggregates were basaltic material with high porosity (Steypustöðin ehf, 2023; Sveinsdottir & Wigum, 2003).

A summary of the aggregates used in this study is shown in Table 1. To distinguish the materials from the general name of aggregate type, their initial letters are written in capital. For instance, crushed brick and crushed volcanic aggregates are referred to as “Brick” and “Vatnsskard”, respectively. Furthermore, the weak and strong granite aggregates are named “Granite36” and “Granite25”, based on their measured Los Angeles abrasion (LA) values, as shown in Table 1. The bulk density of the materials was measured according to ASTM C127 (2015) standard. To estimate the breakage vulnerability of the aggregates used in this study, their LA values are presented in Table 1. However, the LA values for the Brick aggregates were not available. Existing literature reports that LA value of various brick aggregates ranges from 34 to 37% (Arulrajah et al., 2013; Diagne et al., 2015; Ellis et al., 2015). This indicates that Brick may exert crushing resistance similar to Granite36.

Table 1. The properties of the aggregates (Paper III).

Name	Picture	Bulk Density [g/cm^3]	LA Value [%]
Brick		2.07 ± 0.02	34 – 37
Granite25		2.65± 0.06	25
Granite36		2.71 ± 0.02	36
Vatnsskard		2.61 ± 0.07	28

In Table 2 a summary of the materials and loading conditions investigated with confined monotonic uniaxial compression tests is presented in terms of aggregate types and size fractions in UGMs and maximum loading magnitude P_{max} applied in the tests.

In Papers I–III, the UGMs were composed solely of a single aggregate type. In Paper IV, blended UGMs consisting of Brick and Granite25 were also examined. All blended UGMs evaluated in Paper IV had particle sizes ranging from 2 to 16 mm, and their gradation followed the maximum-density line based on the Fuller gradation. To clarify the composition of the blended UGMs, their compositions and naming, are presented in Table 3 to summarize which aggregate types are used in each case. In the table, orange and grey colours denote Brick and Granite25, respectively, used for each sieve size.

Table 2. Aggregate types and UGMs investigated in confined uniaxial compression tests in this study.

	Aggregate type	Aggregate size range in UGMs	Maximum loading magnitude, P_{max}
Paper I	Granite25	2 – 16 & 11.2 – 16	20, 35 & 50
Paper II	Granite25 & Granite36	8 – 11.2 & 11.2 – 16	20
Paper III	Brick, Granite36 & Vatnsskard	4 – 8, 4 – 11.2 & 11.2 – 16	10 & 20
Paper IV	Blends of Brick & Granite25	2 – 16	5, 10, 20 & 35
Paper V	Measurements from Paper I and III		

Table 3. Composition of blended UGMs before tests. Compositions are color-coded, with orange indicating Brick and grey indicating Granite25 (Paper IV).

Sieve Size	Blend 1	Blend 2	Blend 3	Blend 4
11.2 – 16				
8 – 11.2				
4 – 8				
2 – 4				

2.2.2. Confined uniaxial compression test

The compression tests are conducted at room temperature using an MTS servo-hydraulic load frame with a load capacity of 100 kN. The testing procedure, illustrated in Figure 7 and described in detail in Paper I-III, begins with sieving for separating the aggregates into their respective sieve sizes. The aggregates are then uniformly filled into a cylindrical steel mould with a height of 205.00 mm, a diameter of 100 mm, and a wall thickness of 10 mm, as shown in Figure 7. Each specimen is poured in eight identical fillings. Between each filling, the mould is vibrated externally using a mallet to ensure better packing. Once the mould is filled, a cylindrical steel loading plate with a diameter of 100 mm and a height of 24.50 mm is placed onto the UGM, for transferring the axial load to the specimen in the mould. Then, the UGMs are pre-loaded with a small sitting load below 5% of P_{max}. Subsequently, the UGM is compressed in displacement-controlled mode at a rate of 0.03 mm/s, while piston displacement and reaction force are recorded continuously. The axial compression is stopped when the loading magnitude reaches the predefined value of P_{max}. The specimen is then unloaded and sieved again to measure

aggregate breakage. Three repetitions for each gradation are conducted for each maximum loading magnitude tested.

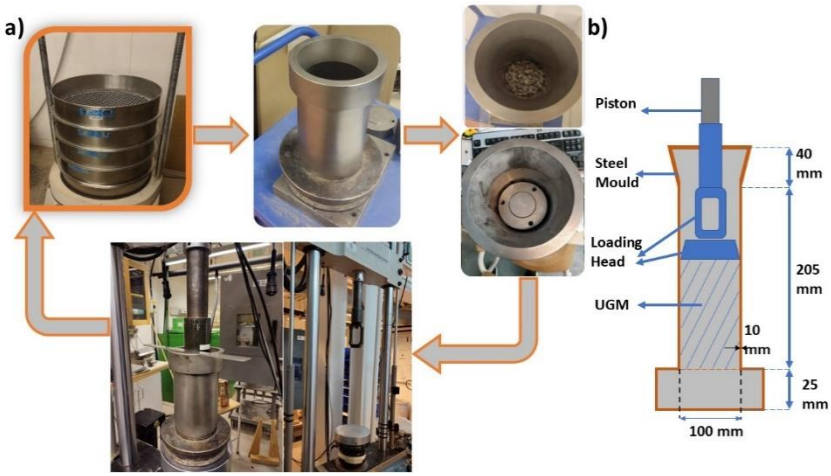


Figure 7. a) Test equipment and procedure schema for confined uniaxial compression test, b) Vertical cut through the cylindrical steel mould along with the loading head (Paper III).

The aggregate breakage is characterized using the so-called Bg-Index, as formulated by Marsal et al. (1965):

$$\mathrm{Bg} = \sum_{i=1}^{M-1} \langle \Delta S_a^i - \Delta S_b^i \rangle, \qquad (22)$$

where M represents the number of sieves used, and ΔS_a^i and ΔS_b^i are the mass of the material measured for each sieve size i after and before the compression, respectively. A schematic illustration for Bg-Index calculation is depicted in Figure 8. Macaulay brackets, <>, indicates that only positive values are included in the summation.

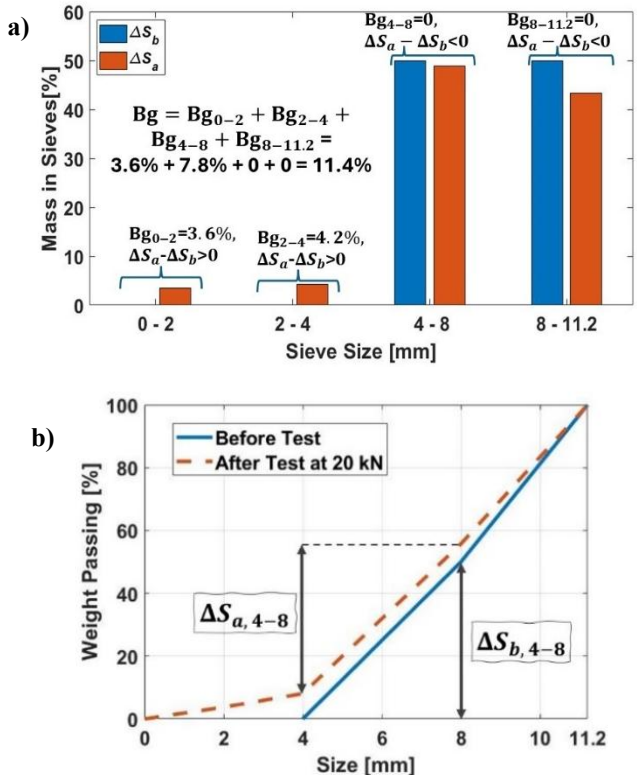


Figure 8. a) An example of Bg-Index calculation, b) A gradation curve of Vatnsskard UGM with a size between 4 and11.2 mm is shown both before and after testing at P_{max} = 20 kN. The definition of the quantity from sieve size 4 - 8 mm is presented and it is utilized for the Bg-Index evaluation in Eq. (22) along with other sieve sizes (Paper III).

2.2.3. Single particle crushing test

In the single particle crushing tests, described in detail in Paper III, aggregates of two sieve sizes, 4 – 8 and 11.2 – 16 mm, are tested. For each size and material, a minimum of 12 aggregates is tested. Aggregates of varying shape, as illustrated in Figure 9, are used to represent those within UGMs.

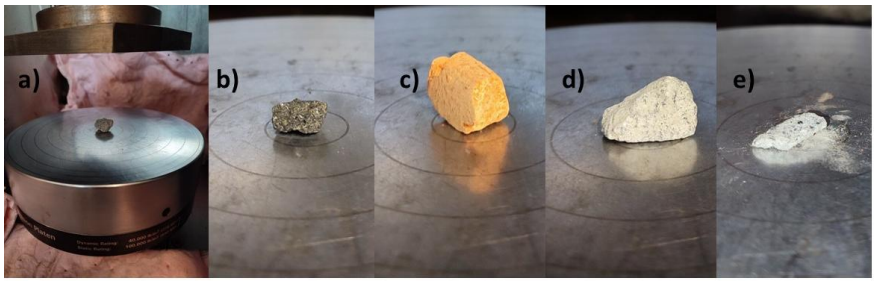


Figure 9. a) Single particle crushing testing setup, b) angular Vatnsskard aggregate (4 – 8 mm), c) Partly rounded Brick aggregate (11.2 – 16 mm), d) Angular Granite25 aggregate (11.2 – 16 mm), e) Elongated Granite36 aggregate (4 – 8 mm) (Paper III).

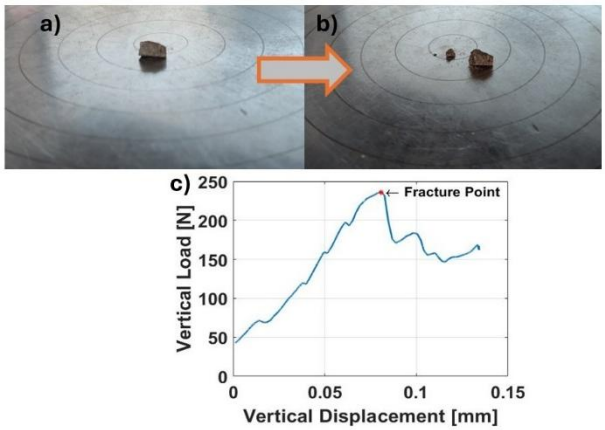


Figure 10. A Granite36 aggregate a) before testing, b) after testing (split into 2 pieces), c) Force-displacement reaction until fracture (Paper III).

The testing procedure is as follows. Initially, the aggregate is placed between two steel plates, as shown in Figure 9(a). The aggregate is preloaded until 40 N, after which it is compressed in displacement-controlled mode with a constant velocity of 0.01 mm/s while displacement and the reaction force are recorded. The critical force value is depicted as the point before the reaction force drops at least 15% of the maximum value, as shown in Figure 10(c).

2.2.4. X-Ray computed tomography measurements and image analysis

X-ray computed tomography (CT), combined with image analysis, is used in Paper IV to evaluate the spatial distribution of fractured aggregates, and fragment size distributions in a blended UGM consisting of Brick and Granite25 aggregates of size 11.2 – 16 mm. The UGM consists of half volume of Brick and Granite25. CT measurements were performed using an NSI X5000 X-ray CT scanner. The specimen was scanned before and after the uniaxial compression tests using identical beam energy and geometric magnification settings. Scanning was conducted at an energy of 225 kV without beam filtration, resulting in a spatial resolution of 71 × 71 × 71 µm.

For CT image acquisition, the UGM specimen was placed in a polyethylene (PE) mould due to its X-ray permeability as shown in Figure 11. The PE mould had a height of 280.0 mm, an inner diameter of 101.0 mm, and a wall thickness of 10.0 mm. To prevent aggregate loss, a cylindrical steel platform with a height of 25 mm and a diameter of 100 mm was placed inside the PE mould. The UGM was then filled into the mould following the procedure described in Subsection 2.2.2.

After initial CT imaging, the PE mould containing the specimen was inserted into a steel mould (diameter 150 mm, height 205 mm, wall thickness 10 mm), as shown in Figure 11(b), with seven PE confinement rings placed between the moulds to restrict lateral movement. Uniaxial compression was then applied in accordance with the procedure outlined in Section 2.2.2, after which the specimen was removed from the confinement and scanned again.

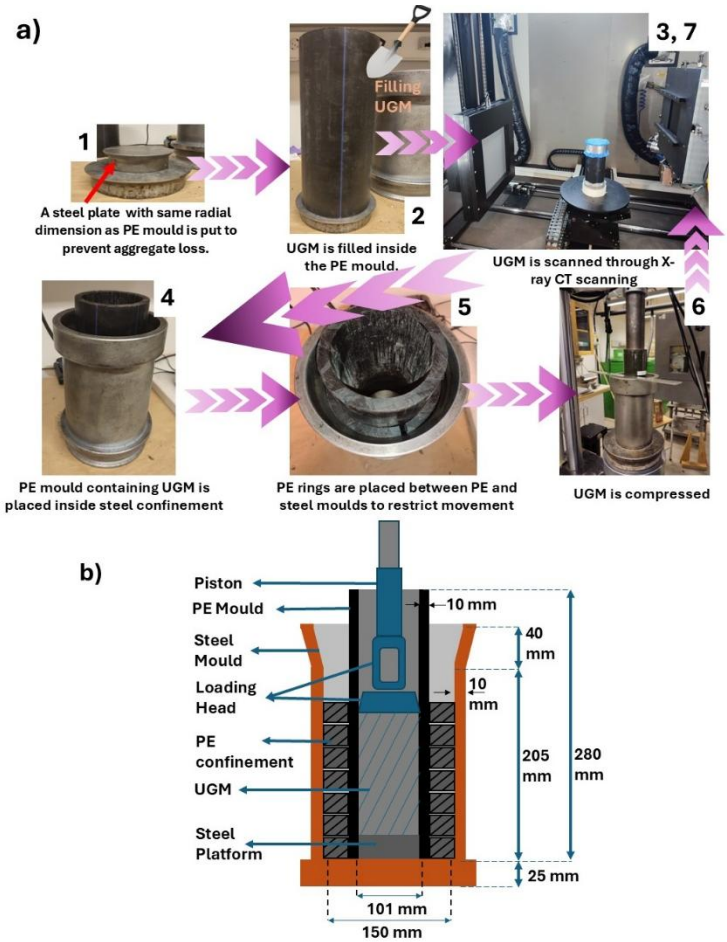


Figure 11. a) The procedure of CT analysis before and after compression, b) The illustration of the cross-section of the moulds. (Paper IV)

The measurements obtained from CT were analysed to identify different aggregate types, their spatial distribution, and their fracture characteristics using commercial image analysis software Avizo 2025.1. An overview of the image analysis procedure is illustrated in Figure 12. An example of an unprocessed CT slice is shown in Figure 12(a). The slice contains three phases: Brick aggregates, Granite25 aggregates, and air voids, which appear as a uniform grey level, variable grey levels due to heterogeneous composition of granite, and dark grey regions, respectively.

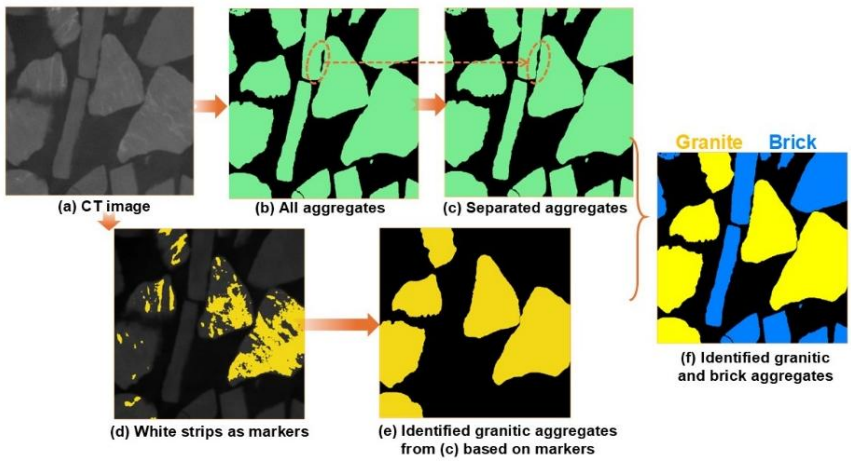


Figure 12.CT image analysis procedure (Paper IV).

The aggregate phase (Brick and Granite25) was segmented using grey-level thresholding, seen in Figure 12(b), followed by separation of individual aggregates using distance-map-based watershed segmentation, shown in Figure 12(c). Granite25 and Brick were subsequently distinguished based on grey-level characteristics, with aggregates associated with high-density regions classified as Granite25 and the remaining aggregates classified as Brick. Aggregate size was characterised using an equivalent sphere approach, where the volume of each aggregate was converted into an equivalent radius assuming spherical geometry.

2.2.5. New parameter to characterize aggregate fracture from compressive test measurements

In Paper V, experimental results from Papers I and III are used to systematically evaluate how UGM parameters influence aggregate fracture across UGMs with different gradations and with both standard- and marginal-quality aggregates. Aggregate strength is additionally characterized using the Los Angeles (LA) abrasion test, and the sensitivity of the LA test and the confined compression test to key material and loading parameters is compared. Statistical analyses are applied to quantify the effects of gradation, aggregate type, and LA abrasion value on aggregate fracture over multiple load levels in uniaxial confined compression tests.

As the Bg-Index, defined by Equation (22), quantifies aggregate fracture only indirectly through changes in the UGM sieve curves, it may not capture the full extent of fracture within the material. This limitation is particularly relevant for broadly graded UGMs, for which the Bg-Index may underestimate breakage. For example, if fragments of a fractured 11.2 mm particle are retained on the 4 mm sieve, their contribution to the Bg-Index depends on whether the original 4 mm particles also fracture and pass to smaller sieves. Moreover, because breakage is assessed by re-sieving the material after testing, tracking the progression of aggregate fracture with increasing load is cumbersome and requires a large number of tests at different maximum load levels.

To address these shortcomings, an alternative approach for characterizing aggregate fracture in UGMs is proposed and evaluated in Paper V. Rather than relying on gradation changes, the method examines differences in the load–displacement response of UGMs in the presence and absence of aggregate fracture, under the hypothesis that these differences are, at least to some extent, proportional to the amount of fracture. The corresponding fracture evolution parameter may thus be defined as:

$$u_B(P_z) = u_{experimental}(P_z) - u_{no\ breakage}(P_z) \tag{23}$$

where $u_{experimental}(P_z)$ shows the measured load–displacement response, and $u_{no\ breakage}(P_z)$ represents the hypothetical load–displacement response of the UGM without aggregate fracture. To capture the $u_{no\ breakage}(P_z)$ function, the DEM model developed in Papers I-III is employed to generate a hypothetical load–displacement response of the UGM in absence of aggregate fracture as schematically shown in Figure 13.

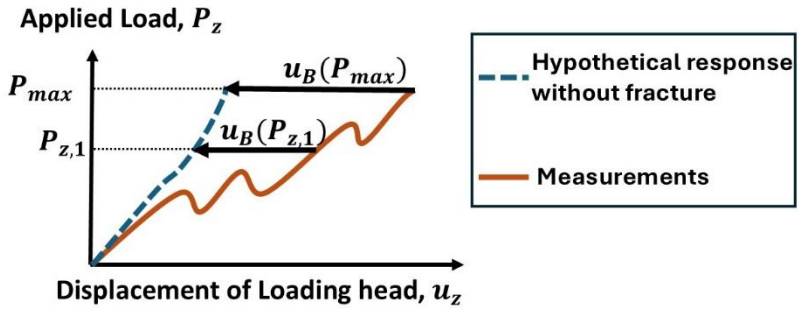


Figure 13. The differences between with and without fracture load-displacement responses to evaluate deformation responses $u_B(P_z)$.

3. Results and Discussion

In this Chapter, the results obtained in this thesis are briefly summarized and discussed. The experimental results of confined uniaxial compression tests on UGMs, obtained in Papers I-IV are presented in Section 3.1, along with the results of X-Ray CT on blended UGMs containing both conventional and marginal-quality aggregates. The DEM results regarding the tests performed on UGMs composed of standard aggregates are presented in Section 3.2. In Section 3.3, results on DEM modelling of aggregate breakage under cyclic loading are presented and the implications of aggregate breakage for macro-scale behaviour of UGM are discussed. In Section 3.4, the results of single particle fracture tests are presented and fracture models, i.e. Models A-C introduced in Subsection 2.1.3, are evaluated with respect to their ability to capture the measurements. DEM modelling results of confined uniaxial compression tests on UGMs containing marginal aggregates, such as Brick, Granite36 and Vatnsskard, presented in Table 1, as well as the blended UGMs introduced in Table 3, are discussed in Section 3.5. Finally, in Section 3.6 a new aggregate fracture evolution parameter, described in Subsection 2.2.5, allowing to capture aggregate fracture from uniaxial confined compressive test measurements is evaluated with respect to its applicability to quantify the progression of aggregate fracture in UGMs.

The UGMs presented in Subsection 2.2.1, are named by starting with the name of the aggregate type, followed by the sieve size range. For example, Granite25 (11.2 – 16) indicates that the UGM is composed solely of Granite25 aggregates, with aggregates passing through a 16 mm sieve and retained on an 11.2 mm sieve, creating a size range of 11.2 – 16 mm.

3.1. Confined Compression Tests on UGMs

3.1.1. UGMs containing standard aggregates

The effects of UGM's gradation and magnitude of applied uniaxial compressive loading on aggregate fracture, were described in Paper I for Granite25 (11.2 – 16) and Granite25 (2 – 16) at P_{max} = 20, 35 and 50 kN.

The experimental results are presented in terms of the applied force at the loading head, P_z, as a function of the piston displacement, u_z, as well as the Bg-Index for quantifying the extend of aggregate fracture. Figure 14(a, b) displays the $P_z(u_z)$ results of Granite25 (11.2 – 16) until P_{max} = 20 kN and Granite25 (2 – 16) until P_{max} = 50 kN. The results of $P_z(u_z)$ until P_{max} = 35 kN for each UGM are omitted for briefness. Three repetitions are conducted for each UGM and P_{max} case.

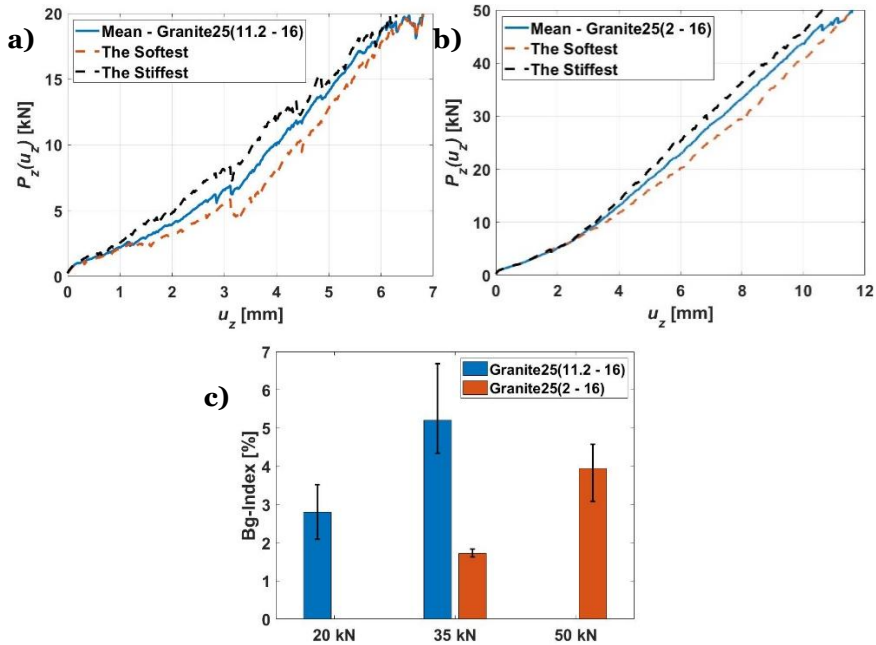


Figure 14.$P_z(u_z)$ graphs of a) Granite25 (11.2 – 16) at P_{max} = 20 kN, and b) Granite25 (2 – 16) at P_{max} = 50 kN from experimental results, and c) measured Bg-Index. Mean of three replicas together with scatter bars showing minimum and maximum values (note that scales are different) (Paper I)

In Figure 14(a, b), the mean values of three repetitions, along with the stiffest and the softest responses, are presented for both Granite25 (11.2 – 16) and Granite25 (2 – 16). Both UGMs demonstrate a similar behaviour and stiffness up to 20 kN. However, noticeable jumps in the $P_z(u_z)$ curve are observed within the final 15 kN and 5 kN ranges for Granite25 (11.2 – 16) and Granite25(2 – 16), respectively. These jumps are attributed to aggregate rearrangements resulting from frictional sliding and breakage. Consequently, significant deviations between the $P_z(u_z)$ curves are observed, particularly for Granite25 (11.2 – 16). Deviations up to 25% from the mean curve are evident for Granite25 (11.2 – 16), whereas deviations for Granite25 (2 – 16) remain below 12% at $P_z = P_{max}$.

In Figure 14(c), the Bg-Index values for each gradation of Granite25 at P_{max} = 20 kN, 35 kN and 50 kN are presented to quantify the aggregate fractures in the UGMs. Each test is represented by the mean of three repetitions, accompanied with the minimum and maximum values. The results indicate that an increase in P_{max} by 15 kN approximately doubles the extend of aggregate fractures for both gradations. Furthermore, the amount of fracture is significantly lower for Granite25 (2 – 16) compared to Granite25 (11.2 – 16). This difference may be associated with the higher number of stone-to-stone contact points in the Granite25 (2 – 16) UGM, which results in reduced contact forces and, consequently, less damage to the aggregates. Additionally, the scatter in the minimum and maximum Bg-Index values, as shown in Figure 14(c) is larger for Granite25 (11.2 – 16) in comparison to Granite25 (2 – 16).

In Paper II, the effect of aggregate type on aggregate fracture in UGMs is evaluated. The study examined Granite25 (11.2 - 16), Granite36 (11.2 – 16) and a blended UGM consisting of two equal layers: namely Granite25 aggregates placed on top of Granite36. This blended UGM is referred to as Granite25/Granite36 and consist of aggregates with sizes between 8 – 11.2 mm. The UGMs are tested up to P_{max} = 20 kN. These UGMs' $P_z(u_z)$ results along with weight passing graph after compression and Bg-Index results are displayed in Figure 15. For each UGM three tests are conducted.

In Figure 15(a), the mean $P_z(u_z)$ lines, alongside the corresponding measurements, are displayed for each UGM. Granite36 (11.2 – 16) exhibits

the most compliant behaviour, whereas Granite25 (11.2 – 16) exhibits the stiffest response.

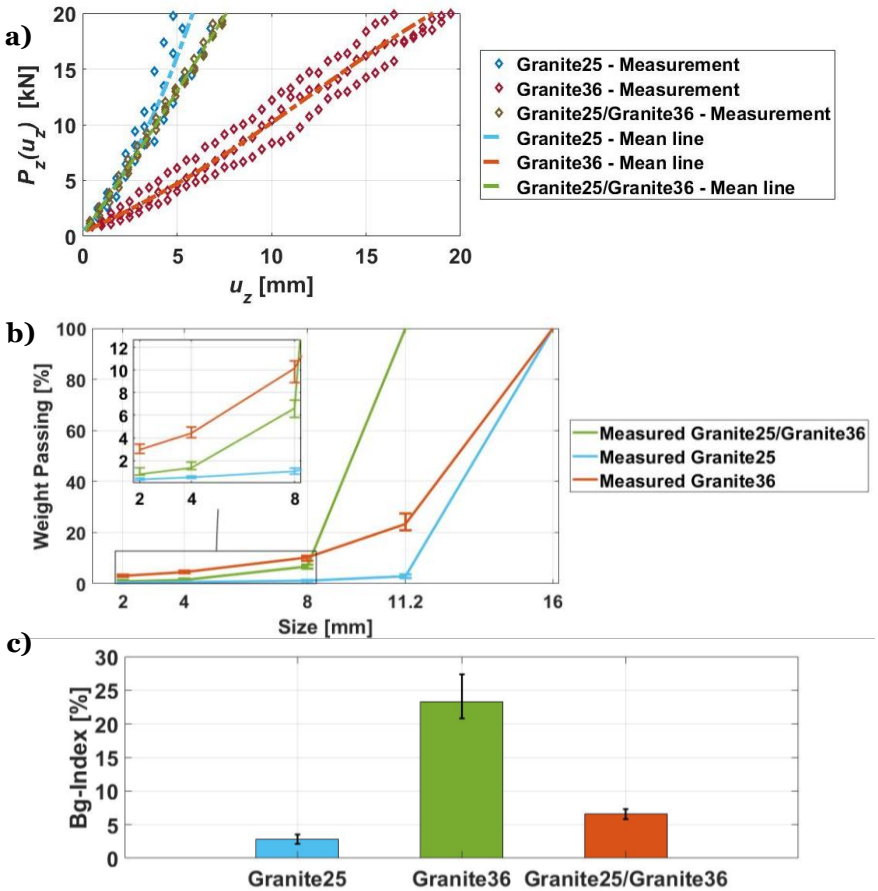


Figure 15. Illustration of Granite25, Granite36 and Granite25/Granite36 at P_{max} = 20 kN for a) $P_z(u_z)$ graph, b) Mean particle size distribution after compression, c) Bg-Index. Mean of three replicas together with scatter bars showing minimum and maximum values. (Paper II)

The mean particle size distributions following compression at P_{max} = 20 kN are illustrated in Figure 15(b). For Granite25 (11.2 – 16), approximately 70% of the fractured aggregates are retained on the 8 – 11 mm sieve, with the remaining aggregates shifting to finer sieve sizes. In contrast, only 50% of the fractured Granite36 (11.2 – 16) aggregates are retained on the 8 – 11 mm sieve, with the remainder passing to smaller sieves. This observation indicates that Granite36 tends to result in smaller fragments than Granite25 after fracturing.

In Figure 15(c), the Bg-Index is presented in a manner similar to that of Figure 14(c). The least amount of aggregate fracture is observed for Granite25 (11.2 – 16), while the highest amount of fracture, along with highest scatter, is observed for Granite36 (11.2 – 16). Additionally, the performance of Granite25/Granite36 is observed to be closer to Granite25(11.2 – 16) than Granite36 (11.2 – 16) in terms of P_z (u_z) behaviour and aggregate fracture as shown in Figure 15(a, c).

Based on the quantitative observations from the particle size distribution in Figure 15(b), a simple method is assumed for evaluating the Bg-Index in DEM. The total weight of a fractured Granite25 particle is assumed to shift to the next sieve. In contrast, for Granite36 particle, only half of the weight is considered to shift to the next sieve after fracture, while the remaining half is assumed to be retained on the same sieve as it was prior to fracturing.

The results presented in Figure 14 reveal significant differences in both P_z (u_z) behaviour and the amount of aggregate fracture across the different gradations and P_{max} values, while Figure 15 highlights these differences for different aggregate types.

3.1.2. UGMs containing marginal strength aggregates

In Paper III, the aggregate fracture performances of UGMs composed of marginal-quality aggregates, i.e. Granite36, Brick and Vatnsskard in Table 1, were examined.

In Figure 16, the mean results from the three test replicates for each UGM are presented, along with the corresponding minimum and maximum values, indicated by variation bars. The Bg-Index varies in a wide range depending on gradation and aggregate type. For instance, Bg-Indicies of 2% and 28% are observed for Vatnsskard (4 – 8) at P_{max}= 10 kN and Granite36 (11.2 – 16) at P_{max} = 20 kN, respectively.

The amount of aggregate fracture increases by approximately three times when P_{max} is increased from 10 to 20 kN for UGMs of 11.2 – 16 mm size, Brick (4 – 8) and Granite36 (4 – 8), while the fractured aggregate for the other UGMs amounts approximately double. Notably, the ranking of materials does not follow a consistent trend across the different size ranges tested. For example, Granite36 (11.2 – 16) exhibits the highest amount of fracture of all

(11.2-16) materials at P_{max} = 10 and 20 kN, whereas Brick UGMs exhibit the highest amount of fracture for 4 – 8 mm and 4 – 11.2 mm size ranges. This variability in ranking can be attributed to the volume and shape dependencies of the material strength, as discussed further in Section 3.4.

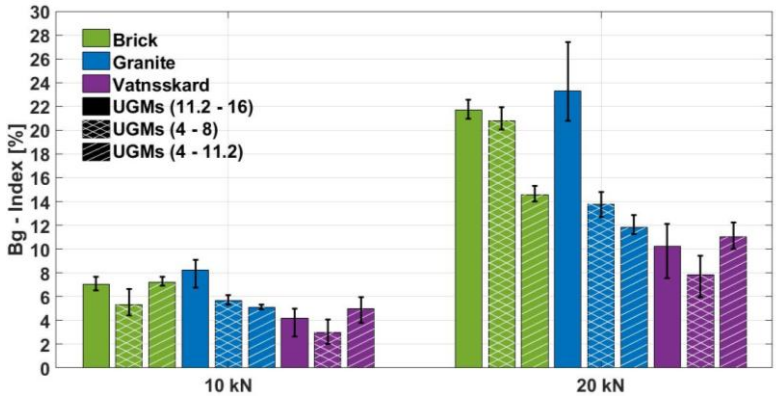


Figure 16. Measured Bg-Index values obtained from mean of three replicates are presented along with their minimum and maximum scatter for UGMs at P_{max} = 10 kN and 20 kN. (Paper III)

In Paper IV, the performance of blended UGMs defined in Table 3 was examined.

Figure 17 presents the results of the uniaxial confined compression tests on blended UGMs, Granite25 UGMs, and Brick UGMs, all having a gradation of 2 – 16. Figure 17(a, b-c) illustrates $P_z(u_z)$ and Bg-Index respectively where all the data are shown as mean values of three repetitions along with scatter of the measurements. At P_z = 20 kN, the corresponding u_z values ranged from 5 – 6 mm for Granite25 UGM, 7 – 11 mm for the Blends, and 21 – 23 mm for Brick UGM, as illustrated in Figure 17(a). Differences in UGM deformability arise from both aggregate type and material porosity, with Brick UGM exhibiting the most compliant behaviour.

All Blend UGMs display considerably greater stiffness than Brick UGM. As observed in Figure 17(a), at P_{max} = 20 kN, deformation of Blends is 40 – 122% higher than that of Granite25 UGM, while Brick UGM exceeds Granite25 UGM by 420%. The type of composition of Blends also influences the UGM stiffness, where a 15% difference between the most compliant blend (Blend 3) and the stiffest blend (Blend 4) is observed. These results suggest that partially substituting conventional road aggregates with marginal materials in UGMs can be an effective strategy for incorporating these

aggregates into road construction without significantly reducing UGM stiffness.

Figure 17(b, c) presents the Bg-Indices measured for all UGMs as total Bg-Index values per UGM shown in Figure 17(b), and for Brick and Granite25 fractions separately, shown in Figure 17(c). Bg-Indices of blended UGMs fall between the values of Brick and Granite25 as can be observed in Figure 17(b). The degree of aggregate fracture in the blended UGMs also depends on their specific composition. Among them, Blend 4 exhibits the lowest Bg-Index, followed by Blend 3, while Blends 1 and 2 display similar fracture magnitude. Comparing the Bg-Indices of the blended UGMs against Brick at $P_{max} = 20$ kN shows that the blended UGMs have 35 – 65% lower Bg-Index values. Notably, the fracture observed in Blend 4 at $P_{max} = 20$ kN is comparable to that of Brick at $P_{max} = 10$ kN, indicating that partial replacement of the gradation with marginal aggregates can effectively reduce aggregate fracture in UGMs. However, the fracture resistance of the blended UGMs remains considerably lower than that of Granite25; for example, the Bg-Index of Blend 4 is roughly 3.5 times higher than that of Granite25 at $P_{max} = 35$ kN. These findings emphasize the importance of identifying load levels so that excessive aggregate fracture does not occur when marginal aggregates are incorporated in pavement construction.

Figure 17(c) shows that fractured Brick amount is consistently higher than of Granite25 within blended UGMs. The magnitude of this difference, however, decreases as the Brick aggregate size decreases. This trend is expected, as coarser aggregates in UGMs are typically exposed to higher contact forces than finer ones. Notably, the Bg-Index of Granite25 fraction within Blend 4 exceeds that of Granite25 UGM at $P_{max} = 35$ kN, indicating that partial substitution of the gradation with marginal aggregates can negatively influence the fracture behaviour of the stronger aggregates.

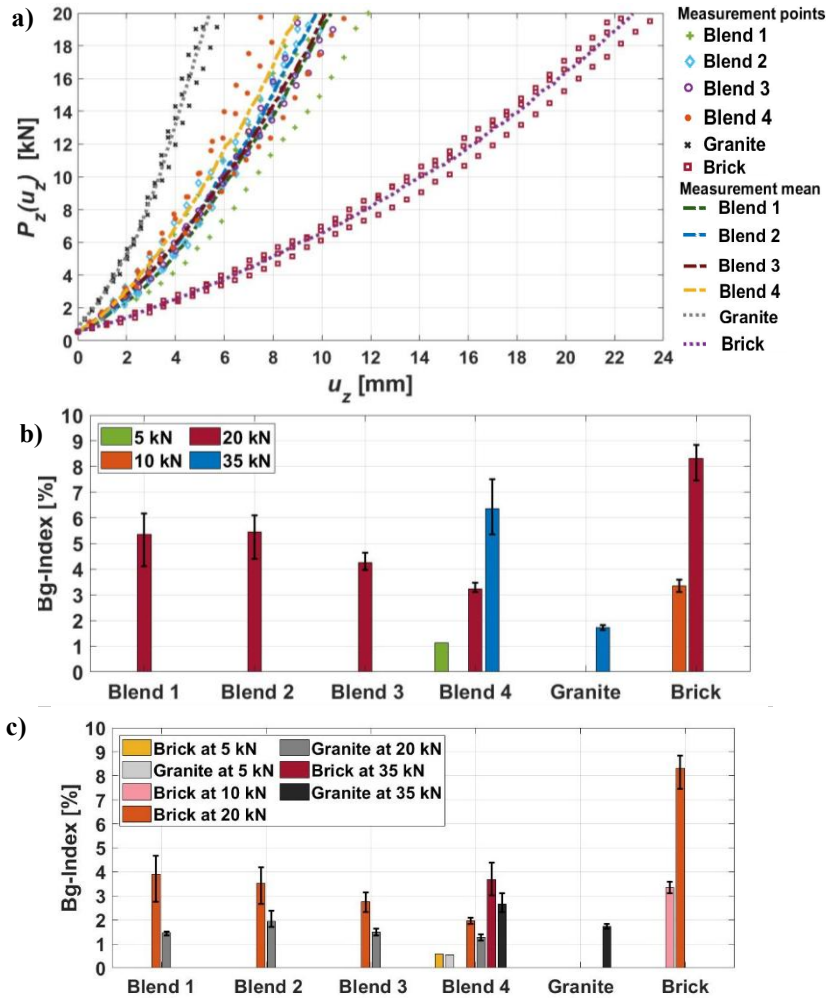


Figure 17. Blend UGMs and pure Granite and Brick UGMs a) measured and mean $P_z(u_z)$, b) Bg-Index where Granite25 and Brick are depicted separately. (Paper IV)

To assess how aggregates fragment after compression, the gradation curves for each aggregate type before and after compression tests are compared in Figure 18(a, b) for Blend 2 UGM. The other blended UGMs exhibit similar behaviour and are therefore not included for brevity. Brick and Granite25 fractions display different fragmentation behaviour. For Brick, approximately 65% of fractured aggregates are retained on the next smaller sieve, while the rest shifts to smaller sieve sizes as shown in Figure 18(a). For

Granite25, however, the changes in gradation are less noticeable in Figure 18(a), due to both limited fracture and the presence of multiple aggregate sizes. On the other hand, Figure 18(b) indicates that the volume lost from the largest Granite25 fraction is nearly equal to the volume retained on the next smaller sieve, a trend also observed for the other Granite25 size fractions. Overall, these observations suggest that Granite25 aggregates tend to break into a few pieces and are retained on the adjacent smaller sieve, whereas Brick aggregates fragment into a wider range of sizes. These results of Granite25 are consistent with the observations in Paper II.

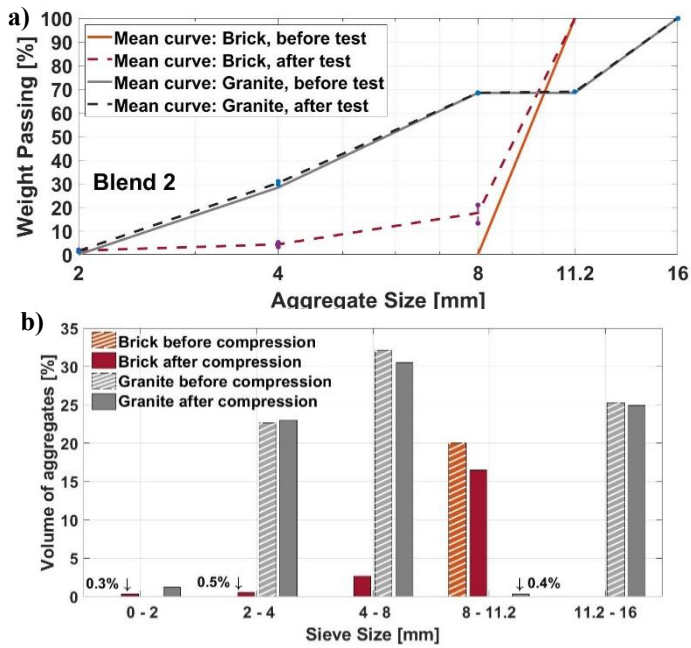


Figure 18. a) Weight passing graph and, b) Volume of retained aggregates in sieves before and after compression in Blend 2. (Paper IV)

3.1.3. X-ray CT image analysis on blended UGMs

In order to evaluate fracture behaviour of aggregates within UGMs, X-ray CT is used, with special focus on assessing fracture patterns, spatial distribution of fractured aggregates, and fragment size distribution, as detailed in Paper IV. The distribution pattern obtained from X-ray computed tomography (CT) observations are then incorporated into the DEM analysis to facilitate comparison between modelling results and measurements.

Aggregate fracture characteristics were examined using X-ray CT on a UGM with a volume composed of 50 Vol-% Granite25 and 50 Vol-% Brick aggregates in the 11.2–16 mm size range, both before and after compression testing, as described in Section 2.2.4. Figure 19 shows the reconstructed surfaces of Granite25 and Brick aggregates obtained from the CT analyses before and after loading. As illustrated in Figure 19(b), fragmentation occurred in several Brick aggregates during compression, with both small and large fragments indicated by red circles.

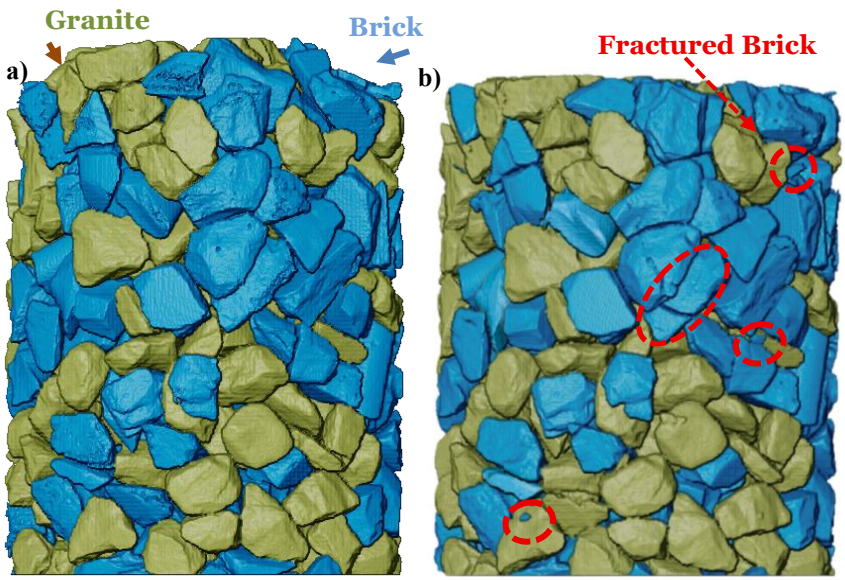


Figure 19. Reconstructed Granite25 - Brick UGM (11.2 - 16) from X-Ray CT: a) before compression, b) after compression. (Paper IV)

Figure 20 presents a comparison between aggregate size distributions obtained from sieve analysis and derived from the corresponding CT measurements before and after loading. In this figure, the volume fraction of

aggregates retained on a given sieve size is expressed relative to the total volume of the corresponding aggregate type: V_{sieve}/V_{total}, where V_{sieve} denotes the volume retained on a given sieve size for the respective aggregate type and V_{total} represents the total volume of that same aggregate type. Complete agreement between the two methods is not expected due to ambiguity in selecting an appropriate linear dimension from CT data to assign aggregates to specific sieve sizes. As shown in Figure 20(a, b), CT-based size distributions include aggregates smaller than 11.2 mm, corresponding to 3% for Granite25 and 18% for Brick aggregates—highlighting the inherent uncertainty in comparing the two methods.

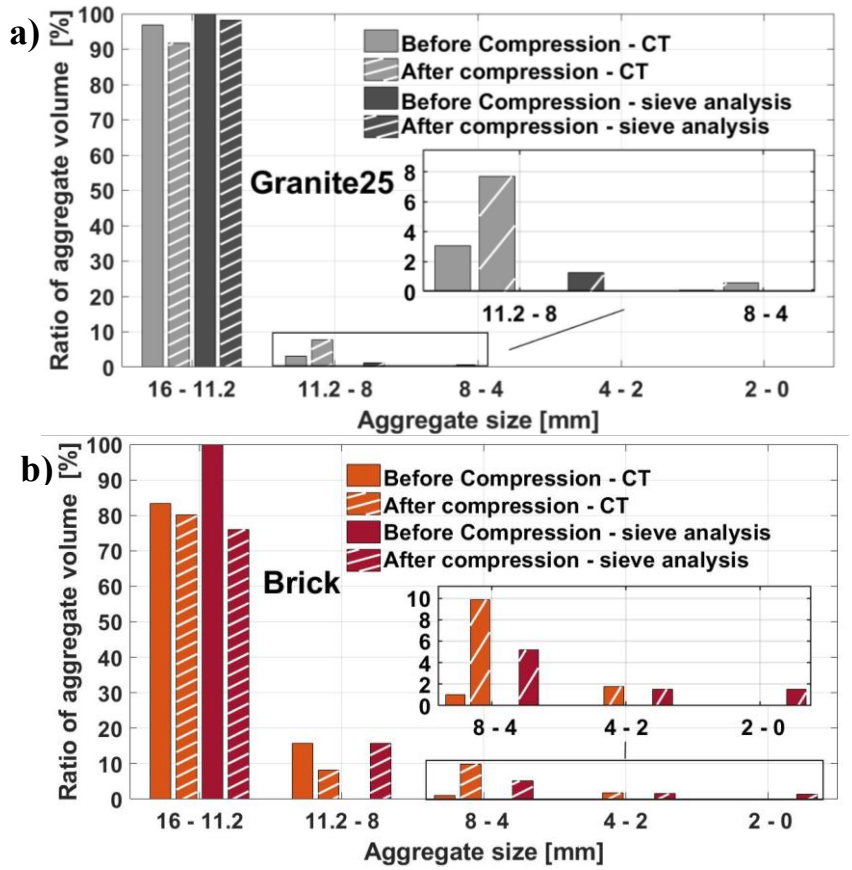


Figure 20. Comparison between measurements and CT image analysis for ratio of Volume retained on sieves: a) Granite25 and b) Brick. (Paper IV)

Nevertheless, the changes in size distribution following compression are observed to be in close agreements between the two methods, as illustrated in Figure 20(a, b). For Granite25, as shown in Figure 20(a), sieve and CT analyses indicates approximately 2% and 5% of the volume is shifted from 11.2 – 16 mm sieve size due to fracture. In both methods, the fractured aggregates are almost entirely moved to next smaller sieve. In contrast, for Brick aggregates, as illustrated in Figure 20(b), approximately 70% of the fractured aggregates from 11.2 – 16 mm sieve size is retained in the next smaller sieve, while the remaining 30% is spread across to smaller fractions.

Figure 21 illustrates the vertical distribution of particles smaller than 11.2 mm in the UGM specimens, used to examine the distribution of fractured aggregates after testing. The data indicate that fractured Brick aggregates are generally distributed approximately uniformly, while Granite25 fragments are mainly concentrated near the bottom of the UGM. The two aggregate types also display distinct fragmentation patterns. All fractured Granite25 aggregates remain larger than 6 mm, whereas Brick fragments span a much broader size range, with some pieces measuring less than 1 mm. Notably, Brick fragments smaller than 3 mm are accumulated near the top of the UGM, while larger pieces are more evenly spread throughout the specimen.

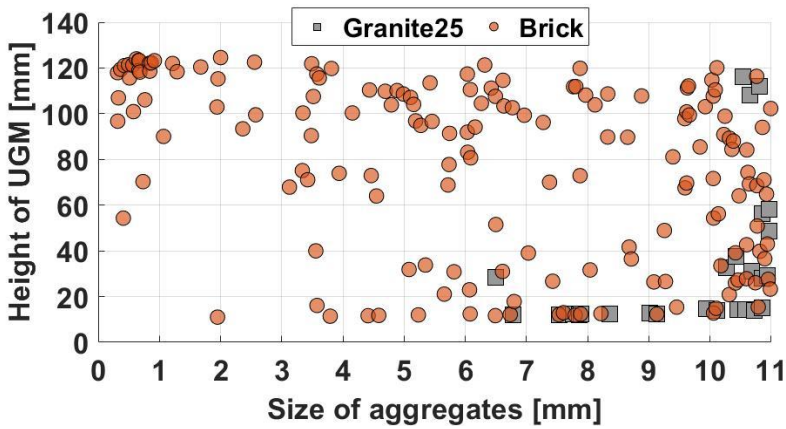


Figure 21. Spatial distribution of finer aggregates (< 11.2 mm). (Paper IV)

3.1.4. Effects of UGM material parameters on aggregate fracture in UGMs

To systematically investigate the influence of UGM material parameters, such as aggregate strength, aggregate type, gradation, and applied load level on the measured Bg-Index, the experimental results obtained in Papers I and III, are evaluated from a statistical point of view in Paper V. As may be seen in Table 2, the cases examined in Papers I and III correspond to a total of 24 different combinations of gradation, aggregate type and applied load magnitude, P_{max}. Three measurement repetitions were performed for each combination, giving a total of 72 data points.

As shown in Paper V, the observed effects of P_{max}, aggregate type and gradation are all significantly larger than the experimental noise. This is illustrated in Figure 22, where each boxplot shows the distribution of absolute changes in Bg-Indicies, either as the standard deviation of the three replicates for a given $aggregate\ type{-}gradation{-}P_{max}$ combination ("replicate noise") or as the change in the mean Bg-Index value per combination, Bg_{mean} ,when only one parameter is varied while the others are kept fixed. As seen in Figure 22, the typical replicate scatter is about 1% Bg-Index**,** whereas changing P_{max} at fixed aggregate and gradation produces median shifts of roughly 6 – 8% Bg-Index**,** and changing aggregate type or gradation produces shifts of about 4 – 6% and 2 – 4% Bg-Index**,** respectively.

The influence of aggregate type, gradation, and load on the amount of aggregate fracture is further quantified in Paper V by fitting a standard ANCOVA-type linear model, cf. Field (2024), to the 24 mean Bg-Index values (one mean per material–gradation–load combination). The predictors used in the ANCOVA model were LA value (LA) and P_{max} as continuous variables, gradation as a categorical factor with four levels (2–16, 4–8, 4–11.2, 11.2–16 mm), and aggregate type as a categorical factor with four levels (Granite25, Granite36, Brick, Vatnsskard). Categorical factor for aggregate type was introduced to capture the potential influence of aggregate properties beyond their strength, which is captured by LA value.

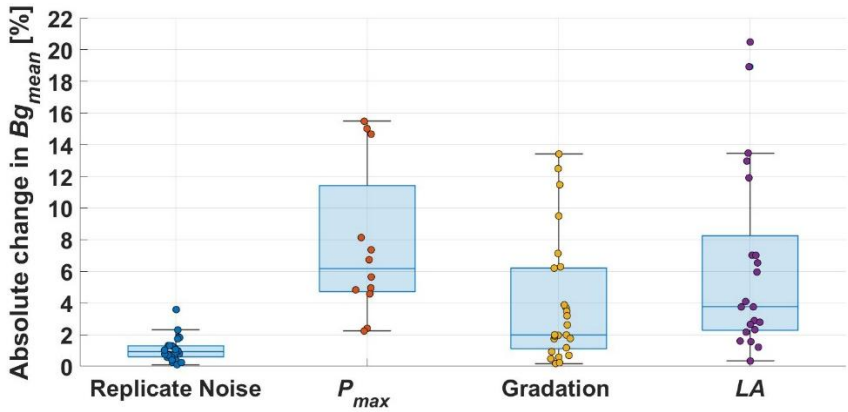


Figure 22.Replicate noise and effects of P_{max}, gradation and aggregate type on absolute change in Bg_{mean}. (Paper V)

The results of ANCOVA analysis are summarized in Table 4. As may be seen, the magnitude of Bg-Index is mainly controlled by LA value and load level: both LA and P_{max} are highly significant (F = 50.7 and 51.7, $p \approx 3\times10^{-6}$ for both) and together account for approximately 65% of the total variance. The interaction term $LA \times P_{\max}$ and gradation also have statistically significance but more moderate effects. Aggregate type, however, when included in addition to LA, contributes only weakly and is not statistically significant at the conventional 5 % level (F = 1.89, p = 0.17, η^2 = 0.04; Table 4). For the present dataset, this suggests that other aggregate parameters (e.g. shape, stiffness) have little additional influence on the measurements beyond that captured by the LA value, although this should not be viewed as a general conclusion.

Table 4. Results of ANCOVA for mean Bg-% (Type II sums of squares).

Effect	F	p-value	Share of total variance (η^2)
LA	50.7	3×10^{-6}	0.32
P_{max}	51.7	3×10^{-6}	0.33
$LA \times P_{\max}$	14.8	0.0016	0.09
Gradation	6.19	0.0060	0.12
Aggregate type	1.89	0.17	0.04
Residual	–	–	0.10

The interplay between LA value, P_{max} and Bg-Index is illustrated further in Figure 23 for UGMs with 11.2–16 mm and 4–8 mm gradations. While there is a general correlation between LA values and Bg-Index, the relationship is not unique, varying significantly with the applied P_{max}. At $P_{\max}$ = 10 kN, the Bg-Index is approximately 1.5 times higher for aggregates with LA = 36% compared to LA =25%. However, at $P_{\max}$=20 kN, the change is more significant, as the Bg-Index difference can become approximately 6 times higher for aggregates with LA = 36% compared to LA = 25%. This substantiates further the finding reported above based on statistical analysis, that the influence of aggregate strength, as represented by LA, on UGM performance depends strongly on the applied load magnitude.

The results in Figure 22 and 23 indicate that reliable prediction of aggregate fracture in UGMs in the field, may not be fully captured based on aggregate tests alone. In addition to aggregate strength characteristics (LA), gradation, applied load magnitude, and their interaction ($LA \times P_{max}$) all exert significant influences on aggregate fracture. Accordingly, accurate predictions concerning the UGM performance with respect to aggregate fracture, requires measurements for realistic gradations and representative load levels.

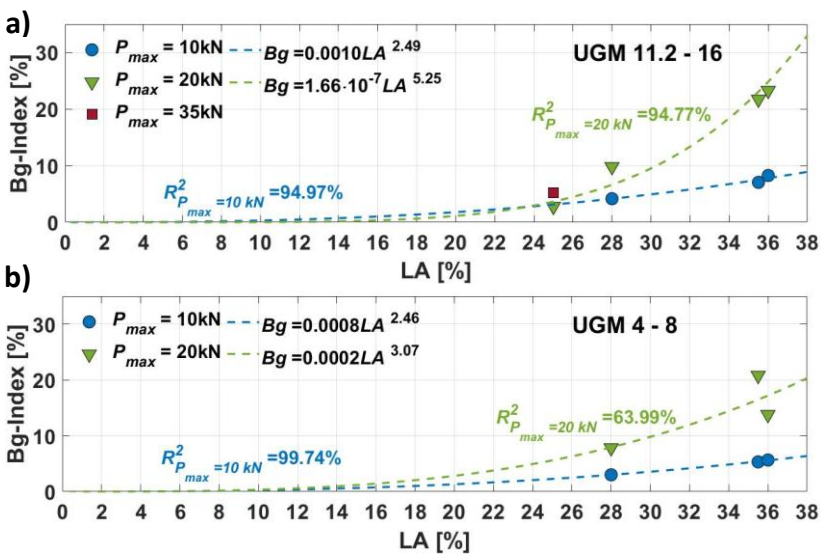


Figure 23. The relationship between Bg-Index and LA values for a) UGMs 11.2 – 16, b) UGMs 4 – 8. (Paper V)

3.2. DEM Modelling of Confined Compression Tests

As discussed in Subections 2.1.2 and 2.1.3 the following parameters are required for the contact laws and fracture model (Model B): Young's modulus, E, friction coefficient, μ, a local damping coefficient, η, and a Poisson's ratio, ν, Weibull stress, σ_B, weakening coefficient, C_W, and Weibull modulus, m_B. Based on the previous studies (Cervera et al., 2017; Olsson et al., 2020), μ, η and ν are fixed, while E, σ_B, C_W, and m_B are determined. In both Papers I and II, the same procedure is used to obtain the parameters and to validate the model: experimental results for one loading magnitude and one gradation are used to identify the parameters while other gradations and loading magnitudes are used for validating. In Paper I, results for Granite25 (11.2 - 16) at P_{max} = 35 kN are used to identify the parameters, and Granite25 (11.2 - 16) at P_{max} = 20 kN, Granite25 (2 - 16) at P_{max} = 35 and 50 kN results are used to validate the model. In Paper II, Granite25 (11.2 - 16) and Granite36 (11.2 - 16) at P_{max} = 20 kN are used for parameter determination, while the Granite25/Granite36 results are used for model validation. The contact law and fracture model parameters identification process is detailed in Paper I.

The simulated P_z (u_z) responses of Granite25 (11.2 – 16) at P_{max} = 20 kN and Granite25 (2 – 16) at P_{max} = 35 and 50 kN are shown in Figure 24, along with the stiffest and the softest curves from the corresponding measurements. In Figure 24(a), the simulation results are found to capture well the measured responses of Granite25 (11.2 – 16). This is expected since the contact and fracture model parameters are obtained from Granite25 (11.2 – 16) at P_{max} = 35 kN. As also seen, the jumps in the P_z (u_z) curve from the simulation show magnitudes similar to those in the measurements, both around 4 kN. This indicates that the effects of aggregate fracture and rearrangement can be captured at macro-scale in the DEM model. As seen in Figure 24(b), the simulation of Granite25 (2 – 16) agrees well with the measured P_z (u_z) responses for $P_z \leq$20 kN. Beyond this point, the simulation deviates and overestimating the stiffness of the UGM at higher loads, with the highest difference in P_z (u_z) curves reaching 20% at P_{max} = 50 kN. This deviation between the measurements and the simulations may be attributed to the modelling of particle fracture, where C_W significantly affects the stiffness of

UGM after particles fracture. This parameter is identified using UGM with larger particles, and it may not accurately capture the effect of particle fracture in smaller particles. On the other hand, the results are encouraging, especially considering that no material parameter is altered between the two gradations. Furthermore, the jumps in $P_z\,(u_z)$ curves for Granite25 (2 – 16) are significantly smaller compared to those in Granite25 (11.2 – 16), leading to a qualitative agreement with the measurements.

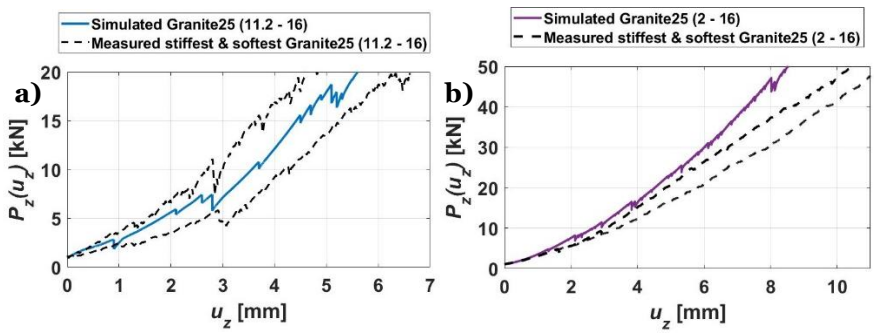


Figure 24.Simulated $P_z\,(u_z)$ curve alongside with measured values for (a) Granite25 (11.2 – 16) at P_{max} = 20 kN and (b) Granite25 (2 – 16) at P_{max} = 50 kN (Paper I).(note that scales are different)

In Figure 25, Bg-Index results for Granite25 (11.2 – 16) and (2 – 16) are presented as a function of P_z, along with the minimum and maximum measured values for P_{max} = 20 kN, 35 kN and 50 kN. The simulation results and measurements exhibit consistent ranking for different loading conditions for each gradation, demonstrating a qualitative agreement between the two. Furthermore, the results of simulations and measurements do not quantitatively deviate much from each other. While modelling results for Granite25 (11.2 – 16) agree with the measurement ranges within the experimental scatter, results for Granite25 (2 - 16) deviate by maximum 0.5% and 1.2% for P_{max} = 35 kN and 50 kN, respectively.

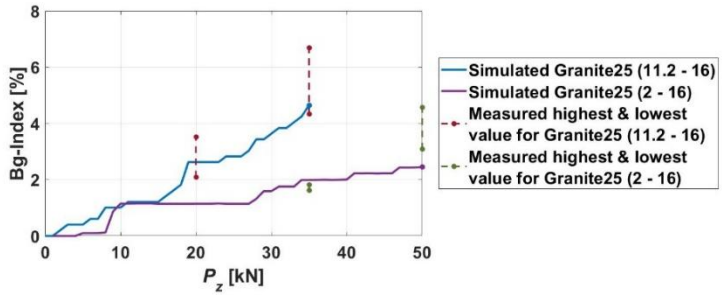


Figure 25. Bg-Index progression for simulated Granite25 UGMs with the measured values. (Paper I)

In Figure 26, the simulated $P_z(u_z)$ responses of Granite25 (11.2 – 16), Granite36 (11.2 – 16) and Granite25/Granite36 are compared to the respective measurements. As shown, the simulation closely matches the measurement up to $P_z \leq 15$kN. However, at higher loading magnitudes, Granite36 (11.2 – 16) and Granite25/Granite36 tend to overestimate the uniaxial stiffness of the UGM. Again, this discrepancy may be attributed to the simplified modelling of particle fracture.

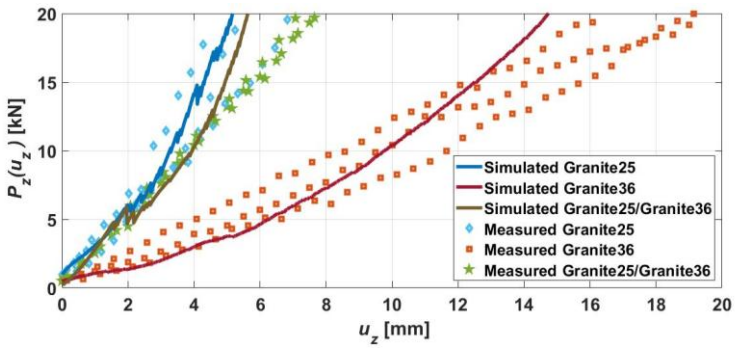


Figure 26. Simulated $P_z(u_z)$ curve alongside with measured values for Granite25 (11.2 – 16), Granite36 (11.2 - 16) and Granite25/Granite36 at P_{max} = 20 kN. (Paper II)

In Figure 27(a), the measured and simulated particle size distributions after the confined uniaxial compressive tests are shown with solid lines and symbols, respectively. The results demonstrate that the model agrees well with the measurements. As shown in Figure 27(a), the difference between the DEM results and the measurements is less than 1% for both Granite25/Granite36 and Granite25 (11.2 – 16). On the other hand, a deviation of up to 6% is observed between the measurements and simulation results for Granite36 (11.2 – 16). This discrepancy may be attributed to the

somewhat simplistic assumption regarding the redistribution of particle weight after fracture.

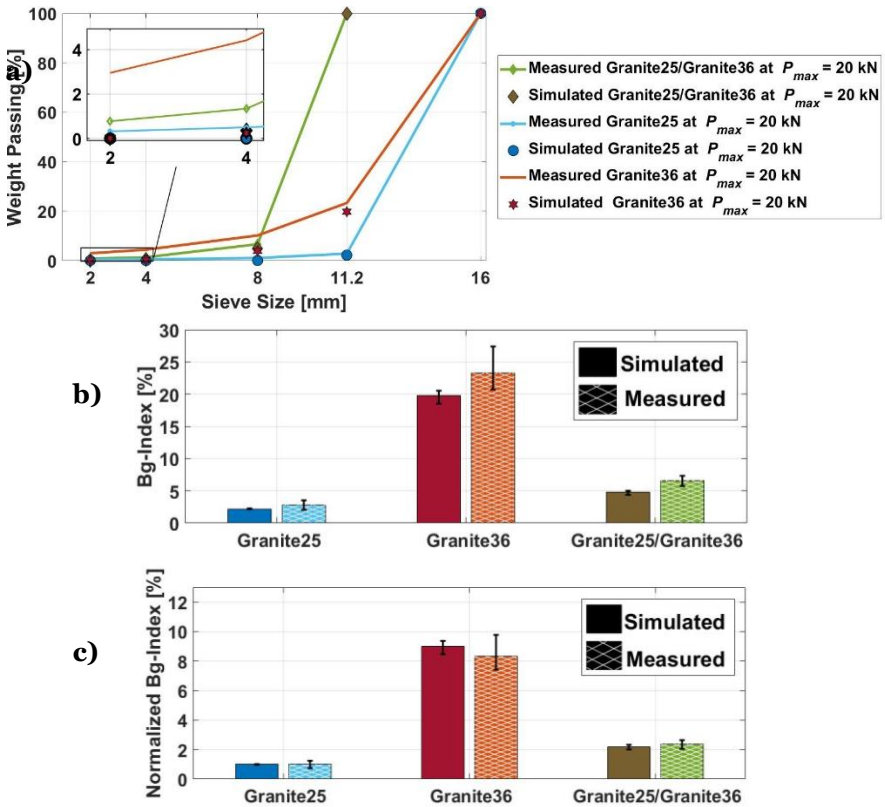


Figure 27. Granite25 (11.2 – 16), Granite36 (11.2 - 16) and Granite25/Granite36 UGM's measured and simulated a) weight passing, b) Bg-Index, c) Normalized Bg-Index. (Paper II)

In Figure 27(b), the experimental and simulated Bg-Index are presented as the mean of three repetitions, with error bars indicating the maximum and minimum values. All computational results agree well with the measurements, deviating by no more than 3%, demonstrating that the model accurately captures the Bg-Index of UGMs. The differences are approximately <1%, 4% and 2% for Granite25 (11.2 – 16), Granite36 (11.2 – 16), and Granite25/Granite36 respectively. In Figure 27(c), the experimental and simulated Bg-Index results are presented normalized to their respective values for Granite25 (11.2 – 16). This normalization emphasizes the relative effect of the aggregate type on UGM fracture performance. As observed in

Figure 27(c), the simulation results are in close match with the normalized experimental data.

In summary, the results presented in Figure 24 – 27 demonstrate that the granular mechanics-based contact law and fracture models, along with their identified parameters, effectively capture the impact of UGM gradations, P_{max} values, and aggregate types on aggregate fracture and $P_z(u_z)$ behaviour, thus validating the modelling methodology to determine these parameters.

DEM, additionally, allows to gain some micromechanical insights into aggregate fracture in UGMs and enables the quantitative assessment of the material parameters that govern this process. In order to illustrate this, the locations of fractured particles are shown along with contact force chains for Granite25 (11.2 – 16), Granite36 (11.2 – 16) and Granite25/Granite36 at P_{max} = 20 kN in Figure 28(a, b, c) respectively. Red and pink spheres represent fractured Granite25 and Granite36 particles, while the magnitude of the contact forces is represented by grayscale intensity and the thickness of the chains, where the lighter, thicker lines represent higher magnitudes. As shown in Figure 28(a, c), contact forces at the top of the specimens generally have higher magnitudes and are less uniformly distributed compared to those at the bottom, since the UGM is being compressed from the top. This difference explains why the fractured particles are primarily localized at the top of the specimen for both Granite25 (11.2 – 16) and Granite36 (11.2 – 16). In the case of the Granite25/Granite36, fractured particles mostly accumulate at the bottom, specifically in the Granite36 layer. However, the number of fractured particles is considerably lower than in Granite36 (11.2 – 16), a trend also reflected in the Bg-Index values presented in Figure 27(b, c) as shown in Paper II and Etikan et al. (2024b).

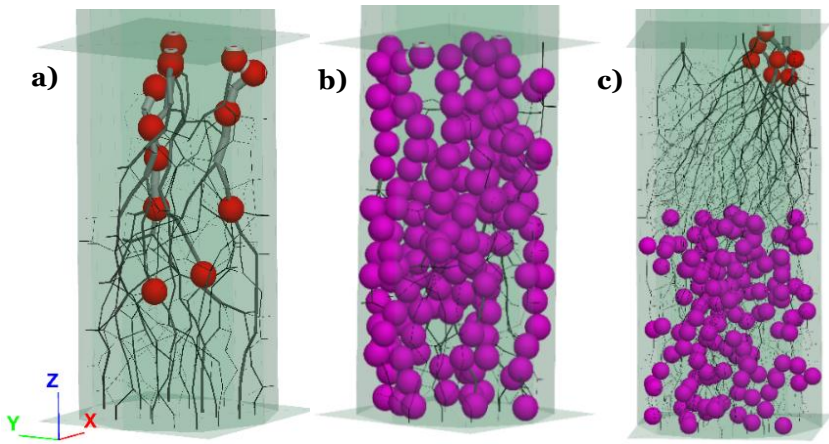


Figure 28. Contact skeleton (where lighter colours indicate higher forces and darker colours indicates smaller forces) with fractured particles (red & pink spheres for granite25 and granite36 materials, respectively) at P_z = 20 kN of a) Granite25 (11.2 – 16), b) Granite36(11.2 – 16), c) Granite25/Granite36 (Paper II).

3.3. DEM Modelling of Aggregate Fracture Under Cyclic Loading

To investigate the impact of aggregate fracture on UGM performance under cyclic loading, different aggregate types and loading magnitudes are analysed in DEM. In Paper II and in Etikan et al. (2024b), UGMs composed of Granite25 and Granite36 are modelled under cyclic loading conditions that simulate the magnitudes encountered at the top and in the middle of unbound granular layer in a typical Swedish low volume road structure with 45mm asphalt on top of 500mm UGM. The detailed information regarding the pavement structure considered is given in Paper II. The stress levels applied to the UGM at the top and in the middle of the unbound granular layers are obtained from low volume simulations conducted in FEM, where a 50 kN load, representing half of the axle load of a truck, is applied to the road. The details of the FEM model are presented in Paper II. The maximum vertical stress, $\sigma_{z,max}$, at the top and in the middle of the unbound granular layers are calculated to be 724 kPa and 183 kPa, respectively.

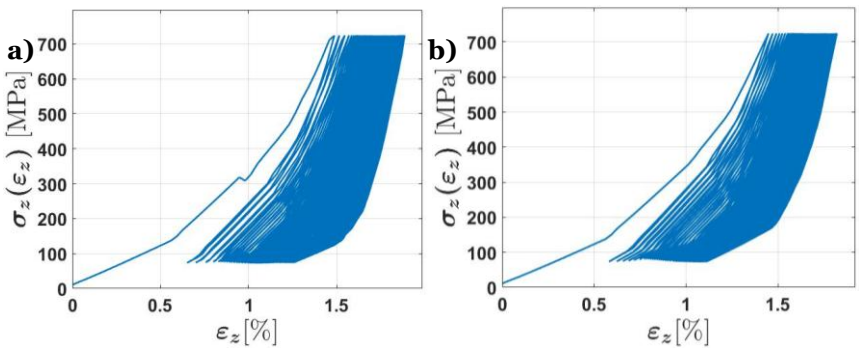


Figure 29.$\sigma_z(\varepsilon_z)$ curves for 100 cycle for Granite36 a) with fracture, b) without fracture (Paper II).

Each UGM is subjected to vertical loading for 100 cycles with a loading rate of 40 mm/s. The UGMs are loaded to the designated $\sigma_{z,max}$ and then unloaded to 10% of $\sigma_{z,max}$. To evaluate the effect of particle fracture on macro-mechanical response of UGMs, comparative simulations are conducted with and without particle fracture taken into account, labelled below as "with fracture" and "without fracture" respectively. To illustrate the effect of particle fracture, the nominal vertical stress, σ_z, is shown as a function of vertical strain of UGM, ε_z, for $\sigma_{z,max}$ = 724 kPa, in Figure 29(a, b) for Granite36 UGM, with and without fracture, respectively. Both cases display similar behaviour, where the UGMs become stiffer and accumulate a comparable amount of permanent strain, $\varepsilon_{z,p}$, during the applied load cycles. It is observed that approximately 40% of $\varepsilon_{z,p}$ accumulates after the first cycle, with the accumulation rate decreasing drastically thereafter. A distinct jump in σ_z is observed in the first cycle of compression, as shown in Figure 29(a). Furthermore, particle fracture leads to higher accumulation of $\varepsilon_{z,p}$ for UGM as depicted also in Figure 30.

In Figure 30(a), the accumulation of fractured particles over load cycles, N, for $\sigma_{z,max}$ = 183 kPa and 724 kPa of Granite25 and Granite36 is illustrated. The ratios of fractured particles are calculated by dividing the number of fractured particles by the total number of particles in simulated UGMs. As shown in Figure 30(a), Granite25 UGM does not exhibit any particle fracture in either case. In contrast, Granite36 UGM accumulates some particle fracture, which reaches a maximum of 3% fractured particle ratio after 100 cycles. The amount of fracture for $\sigma_{z,max}$ = 724 kPa is 8 times higher than for

$\sigma_{Z,max}$ = 183 kPa. These results suggest that placing Granite36 deeper within the unbound layer, rather than in the middle, may be an effective strategy for minimizing aggregate fractures.

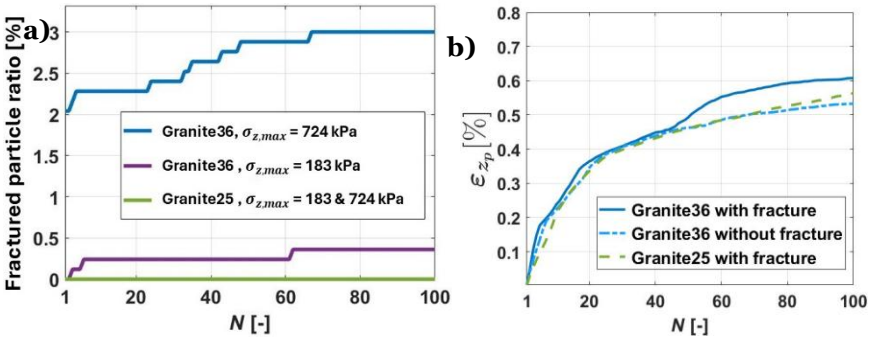


Figure 30. a) Particle fracture evolution over cycles for each material at both layers (Models with fracture are illustrated), b) Accumulation of $\varepsilon_{z,p}$ over cycles for each material at $\sigma_{z,max}$ = 724 kPa. (Paper II)

In Figure 30(b), the effect of particle breakage on the accumulation of $\varepsilon_{z,p}$ is shown for $\sigma_{Z,max}$ = 724 kPa. In this analysis, $\varepsilon_{z,p}$ is set to zero after the first loading cycle to negate the settlement caused by the initial cycles. As shown in Figure 30(b), Granite25 and Granite36 UGMs exhibit nearly identical permanent deformation when aggregate fracture is disregarded. However, when fractures are accounted for, the accumulation of $\varepsilon_{z,p}$ increases by 10% for Granite36 UGM. This higher permanent deformation may promote rutting and contribute to fatigue distress due to the compromised support of the asphalt layers.

The results presented in Figure 29 – 30, illustrate the usefulness of DEM in capturing the effects of micro-mechanical phenomena, such as aggregate fracture, on macro-scale response of UGMs. Accordingly, the DEM modelling may potentially become a useful tool for optimizing UGMs in road construction, as it allows to quantify the effects of road structural design, UGM and aggregate properties on aggregate fracture, thus gaining insight into the influence of aggregate fracture on the macroscopic behaviour of UGMs.

3.4. Single Particle Fracture Tests and Extended Fracture Model

While the results presented in Sections 3.2 - 3.3 show that the developed model captures well the effects of gradation and loading magnitude on the aggregate fracture performance in UGMs under confined monotonic and cyclic compressive loads, some uncertainties still remain. The fracture model used in Papers I and II (Model B) considers only the volume dependency of aggregate strength, which may vary across different aggregate types and will be inaccurate if there is a large variability in aggregate shape. To obtain an accurate description for a wide range of aggregate types, such as aggregates from marginal materials, a new fracture model is needed accounting both for the variability of fracture forces due to aggregate shape and random defects within the aggregate. Such a model is particularly important when accounting for the fracture strength of aggregates of different sizes while optimizing the choice of gradations of marginal aggregates.

In order to overcome these limitations, Paper III proposes a new particle fracture model, cf. Model C in Section 2.1.3, which is evaluated along the Weibull distribution and Weibull weakest link approach for four different aggregate types: Brick, Granite25, Granite36, and Vatnsskard.

The critical force, F_f, distributions, measured in single particle crushing tests introduced in Subsection 2.2.3, are represented as critical nominal stress, F_f/R^2, alongside the critical stress levels predicted by Model A – C in Figure 31, where R^2 is the square of mean radius of the sieve size and calculated as $\left(\frac{R_N+R_{N+1}}{2}\right)^2$. The experimental fracture probability is estimated through the median ranks (Benard & Bos-Levenbach, 1955), where the cumulative fracture probability $p_f^{(i)}$ of tested aggregate i is expressed as:

$$p_f^{(i)} = \frac{i - 0.3}{M + 0.4}, \tag{24}$$

where M denotes the number of tests conducted for a given material and aggregate size. The parameters for the fracture Models A—C are identified through least squares fitting of the experimental data shown in Figure 31. The

resulting parameter, along with the coefficients of determination ($Corr$) are provided in Table 5 – 7.

As can observed in Figure 31, the aggregate size and type notably influence the critical stress of the aggregates. Specifically, at a 50% fracture probability, i.e. $S(F_f/R^2) = 0.5$ the stress values vary between 13 MPa and 81 MPa for the four aggregate types with the sieve size of 11.2 – 16 mm. For the 4 – 8 mm size range, the corresponding values are 19 MPa to 110 MPa, indicating that finer particles display higher critical stress values. Accordingly, a wide range of strength properties can be observed in the aggregates used in this study.

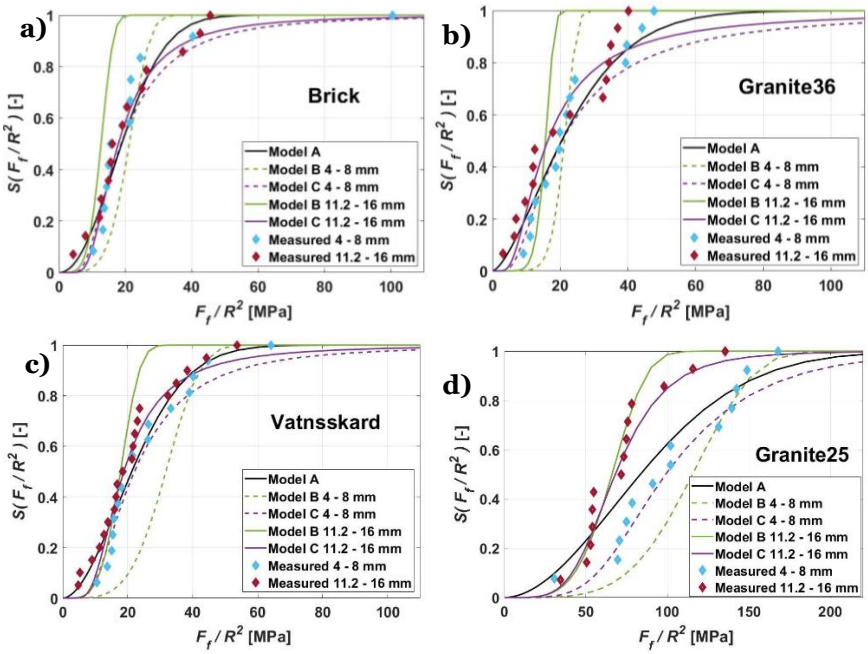


Figure 31. The measurement points are plotted against $S(F_f/R^2)$ for three fracture models, for a) Brick, b) Granite, c) Vatnsskard, d) Granite25. The measurements for the size of 4 – 8 mm and 11.2 – 16 mm are represented by blue and red diamonds, respectively. The fits for the small and large sizes are shown as dashed and solid lines (Paper III). Note x axis has different scales. $S(F_f/R^2)$ for Model A, B and C are calculated through Eqs. (11), (15) and (20) respectively

As shown in Figure 31, Models A and C provide a reasonably good fit to the experimental data for Brick, Granite36 and Vatnsskard, whereas Model B exhibits significant discrepancies. In particular, Model B underestimates the fracture resistance of coarser particles while overestimating the strength of finer particles. Model A does not account for particle size effects on fracture

strength, resulting in a single line for both sieve sizes tested, whereas Models B and C produce distinct lines for the two sieve sizes. On the other hand, the measurements for Granite25 displays a better agreement with Models B and C. This can be attributed to the fact that the strength of Granite25 shows a stronger volume dependency than the other three aggregate types.

Table 5 – 7 present the correlation values, $Corr$, between the fracture models and the measurements. Only Model C yields $Corr > 0.9$ for all the aggregate types tested. Both Models A and C exhibits high $Corr$ values for Brick, Granite36 and Vatnsskard, with Model C slightly outperforming Model A. However, only Model C achieves $Corr > 0.9$ for Granite25, while Model A shows the lowest correlation. The difference in these fittings arises from the capability of each fracture model to account for the strength of particles with varying shape, which is reflected in the scatter of the measured F_f/R^2 data, as seen in Figure 31(a – c). On the other hand, Model B only considers size differences and fails to account for scatter caused by variations in particle shape, resulting in the lowest scatter for Brick, Granite36 and Vatnsskard, except for Granite25. Consequently, both aggregate shape and volume significantly affect fracture forces of the aggregates, and Model C captures both factors, exhibiting the best agreement across all the aggregate types that are tested.

Table 5. Fracture model parameters obtained for Model A.

Material	σ_A **[MPa]**	m_A [−]	$Corr$ [−]
Brick	22.5	2.11	0.937
Granite36	26.0	1.52	0.930
Vatnsskard	25.9	1.85	0.943
Granite25	101.16	1.90	0.745

Table 6. Fracture model parameters obtained for Model B.

Material	σ_B [MPa]	m_B [−]	$Corr$ [−]
Brick	14.5	4.79	0.225
Granite36	16.9	7.99	0.428
Vatnsskard	20.59	4.24	0.538
Granite25	76.0	4.40	0.821

Table 7. Fracture model parameters obtained for Model C.

Material	σ_C [MPa]	m_C [−]	n_C [−]	$Corr$ [−]
Brick	14.5	11.5	2.51	0.953
Granite36	15.5	6.51	1.67	0.931
Vatnsskard	18.7	7.81	2.43	0.976
Granite25	75.9	4.69	5.07	0.958

3.5. DEM Modelling of UGMs Containing Marginal-Quality Aggregates

In order to evaluate the performance of the developed Model C in predicting aggregate fracture within the UGMs composed of aggregates from marginal materials, in Paper III, computational predictions for confined uniaxial compression tests are compared to experimental findings for the following cases: UGMs with gradations of 11.2 – 16, 4 – 11.2 and 4 – 8 composed of Brick, Granite36 and Vatnsskard, at P_{max} = 10 and 20 kN. The performance of Model C is assessed by comparison with Model A, which also exhibits a strong correlation with the measured aggregate fracture strength, as demonstrated in Table 5.

In terms of aggregate fracture quantification, experimental and modelling approaches differ. In experiments, difference in mass for aggregates retained at each sieve before and after compression are measured. In contrast, in the modelling, the number of fractured particles is quantified, since detailed information on how and to what shapes the aggregates fragment after fracture

is not available. To quantify the Bg-Index in DEM modelling, an additional assumption about fragmentation is required. It may however be stated that, provided that the model captures the mechanics of aggregate fracture in UGMs accurately, a unique relationship should exist between the Bg-Index in experiments and the number of cracked particles in the model. Tentatively, a power law relationship may be postulated as follows:

$$N_{fractured} = A \cdot Bg^{b}\,, \tag{25}$$

where A and b are fitting parameters that vary for each material, with Bg representing the measured Bg-Index values, and $N_{fractured}$ denotes the particle fracture ratio defined as number of fractured particles to the total number of particles in the simulations. This fitting employs the least squares method, and the resulting equations are displayed in Figure 32. This approach facilitates quantification of how well the different fracture models yield a unique relationship between the experimentally determined Bg-Index and the simulated number of fractured particles. To assess this, the ratio of fractured particles obtained from the DEM models for all UGMs at P_{max} = 10 kN and 20 kN is presented in Figure 32(a, b ,c) for the Brick, Granite36 and Vatnsskard UGMs, respectively. The results in Figure 32 are plotted as a function of the measured Bg-Index for the corresponding UGM – P_{max} combinations. Each data point on the graphs represents the mean value obtained from three simulations and tests, with interval bars indicating the minimum and maximum values for each material. Additionally, the power law functions are fitted to the data according to Eq. (25) as shown in the figures.

A clear correlation exists between the experimental Bg-Index and the number of fractured particles predicted by DEM models for both Model A and C, as illustrated in Figure 32. The number of fractured particles predicted by Model A is approximately 10% higher compared to Model C. Additionally, coefficients of determination ($Corr$) of the power law fits is higher for Model C, supporting the presence of a unique relationship between the number of fractured particles in DEM simulations and the experimentally measured Bg-Index. In fact, the power law curves derived for Model C align with the measured Bg-Index within the experimental scatter for 16 out of 18 test cases. Model A captures the relationship between the fractured particle ratio and

Bg-Index with a $Corr$ of approximately 90% for Brick and Vatnsskard UGMs. However, this agreement is smaller for Granite36 UGMs, with $Corr$ of 60%. This discrepancy can be attributed to higher sensitivity of Granite36 to volume effects, as this material has the lowest m_C value among the other marginal materials studied. Since Model A does not account for particle volume effects in the aggregate fracture forces, F_f, distribution, it is unable to capture the volume dependency of strength for granite aggregates well.

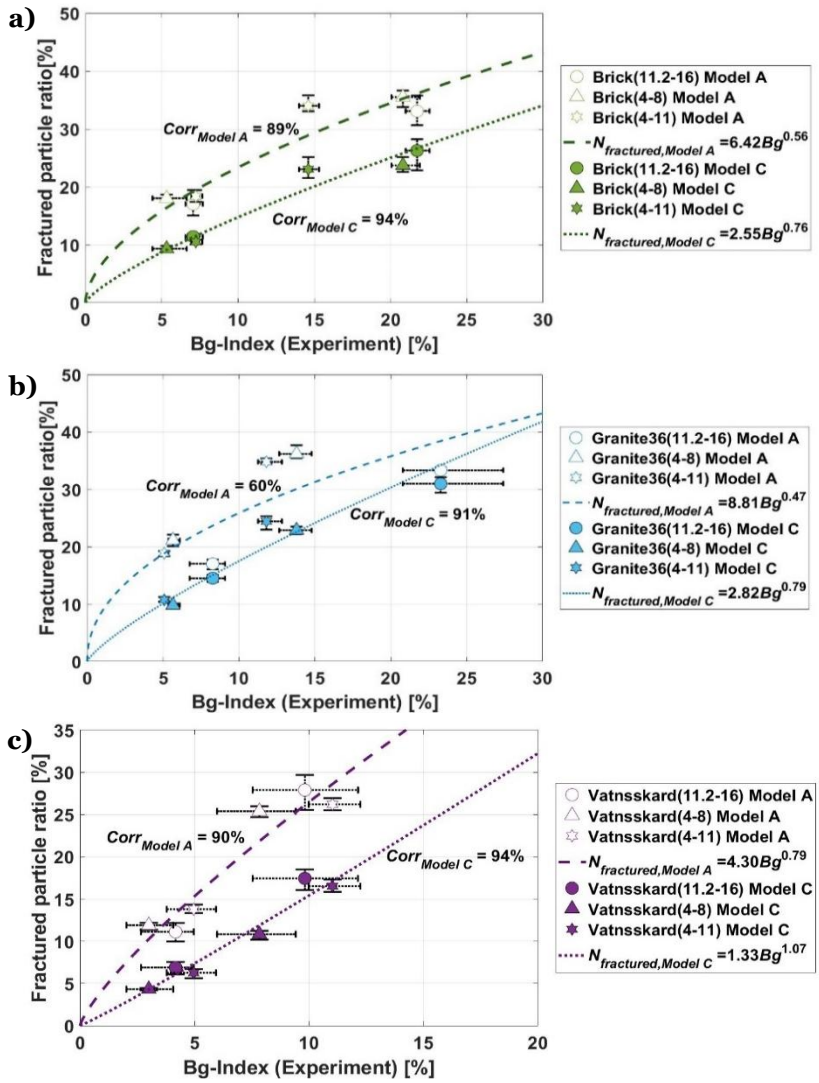


Figure 32. Fracture amount obtained through simulations vs. Bg-Index for a) Brick, b) Granite36, c) Vatnsskard UGMs (Paper III). (note different scales)

The performance of Models A and C in predicting Bg-Index is assessed for all materials in Figure 33. This is achieved by reversing the equations for $N_{fractured,Model\ A}$ and $N_{fractured,Model\ C}$, as defined in Eq. (26), and expressing Bg-Index as a function of the number of fractured particles generated by the DEM model:

$$Bg_{calculated} = (N_{fractured}/A)^{1/b}, \qquad (26)$$

As shown in Figure 33, the Bg-Indexes calculated using Model C exhibit a better match with experimental data than those predicted by Model A. Specifically, $Corr$ = 94 % for Model C and $Corr$ = 74 % for the Model A when compared to the equality line. For Model C, only three data points exhibit a relative error greater than 10%, while for Model A, seven points are observed to fall outside this threshold. The maximum absolute deviations between the calculated and measured Bg-Indexes are 3% for Model C and 7% for Model A.

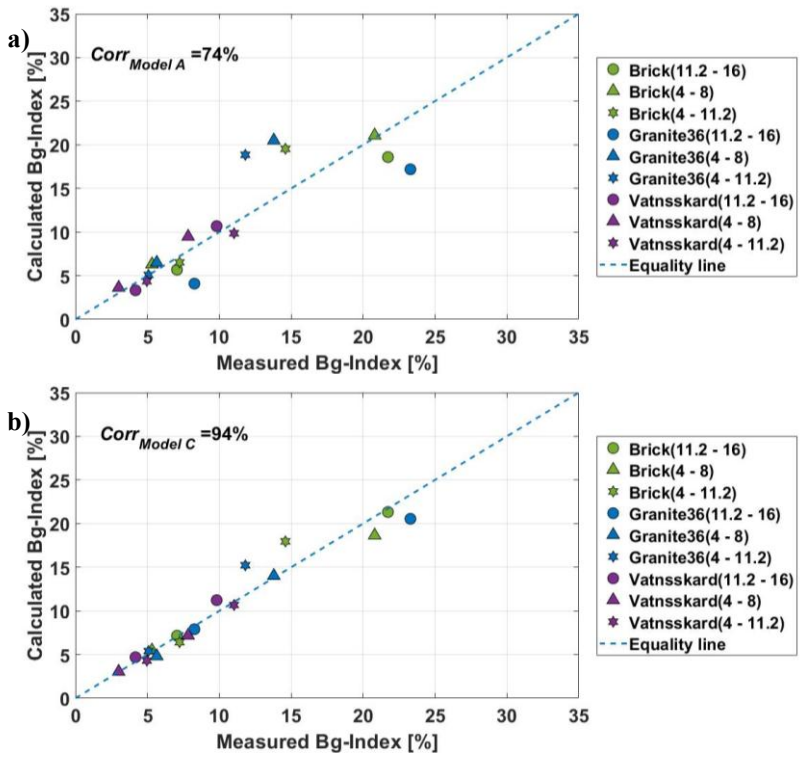


Figure 33. The comparison between the calculated Bg-Index obtained from Eq. (25) and measured BG-Index for a) Model A and b) Model C (Paper III).

Based on the results presented in Figure 32 – 33, it can be concluded that Model C effectively captures the aggregate fracture behaviour across a broad range of UGMs, including variations in aggregate types and gradations. Therefore, it is concluded that an aggregate fracture model that accounts for shape variability and loaded volume size, such as the proposed Model C, is essential for accurately representing aggregate damage in unbound layers.

In Paper IV, the developed DEM framework incorporating Model C is applied to evaluate aggregate fracture in blended UGMs, containing both Granite25 and Brick aggregates, as defined in Table 3. The particular focus in Paper IV, lies on evaluation of DEM capability to capture the effects of UGM composition and of applied load magnitude on aggregate fracture. Accordingly, DEM feasibility in reducing the experimental effort, required for material optimization is evaluated.

In order to compare quantitatively the DEM results in terms of number of fractured particles and measured gradation changes, characterized through Bg-Indices, the following assumption is made regarding how the volume of fractured aggregates redistribute after fracture. Based on the observations from Figure 18 and 20, 65% of the volume of fractured Brick particles is assigned to the next smaller sieve, and of the remaining 35%, 65% is transferred to the subsequent smaller sieve. This iterative redistribution continues for each sieve down to the 2 mm size. In contrast, the total volume of fractured Granite25 particles is assumed to move entirely to the next smaller sieve after fracture. To demonstrate the capabilities of the DEM model in capturing experimental results, Figure 34 compares DEM predictions and measurements of UGM deformation and Bg-Index at $P_{\max}$. The results presented show the mean values of three replicates for both the DEM predictions and the measurements.

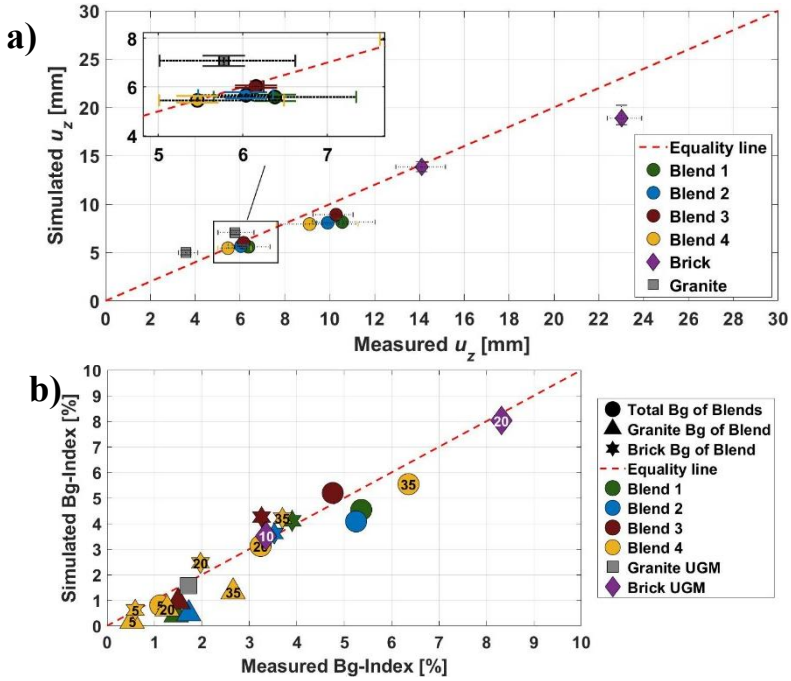


Figure 34. Comparison between DEM and measured responses of Blend UGMs and Granite25 (2 - 16) and Brick (2 -16): a) u_z(P_{max} =10 and 20 kN), b) Bg-Index for all cases along with an equality line. UGMs evaluated at more than one at P_{max} value is marked with the respective P_{max} values. (Paper IV)

Figure 34(a) presents, vertical displacement of the load head, u_z, at P_{max} = 10 and 20 kN predicted by the DEM models as a function of the corresponding measured values, including the scatter, to compare deformations. Except for Brick UGM at P_{max} = 20 kN, all points stand very close to the equality line, indicating that the simulations accurately capture the examined UGM response well up to the force level where aggregate fragmentation becomes significant.

The Bg-Indices of all UGMs examined in Paper IV are compared to the DEM predictions in Figure 34(b) across all load levels evaluated. As may be seen, the simulations match the measurements closely, with differences between mean values below 1.3%, while the maximum deviation, including scatter, reaches 2.4%. The relative ranking of the materials is also well captured. The scatter is omitted, as some values fall below 1% and are difficult to display clearly. More details regarding observed experimental and modelling scatter can be found in Paper IV. The coefficient of determination

between simulation results and the equality line is 90%. The results in Figure 34 are encouraging, indicating that both stiffness and aggregate fracture performance of UGMs are captured with reasonable accuracy. It must be emphasized, that no further calibration of the DEM model has been performed in Paper IV, with all the simulation parameters being identical to Paper III.

In order to illustrate the capability of the developed models to capture the effect of applied load magnitude on aggregate fracture, the evolution of Bg-Index under applied loading from DEM simulations for Blends 1 and 4 along with Brick and Granite25 UGMs are presented in Figure 35 as a function of P_z, alongside the corresponding measured values. The simulations capture the response of all UGMs well, and it is particularly notable that the Bg-Index of Blend 4 is accurately reproduced across all three experimental maximum load levels. The results also show that the influence of load magnitude on Bg-Indices is non-linear and depends on material composition. At loads below 11 kN, Blends 1 and 4 display nearly identical fracture behaviour, with Blend 1 even showing slightly better performance than Blend 4 below 3 kN, whereas at higher loads Blend 4 clearly exhibits superior resistance to fracture.

These observations highlight the importance of assessing aggregate fracture in UGMs at load levels relevant to their intended application. Furthermore, the DEM model demonstrates strong potential as a predictive tool for optimizing blend compositions based on expected loading conditions.

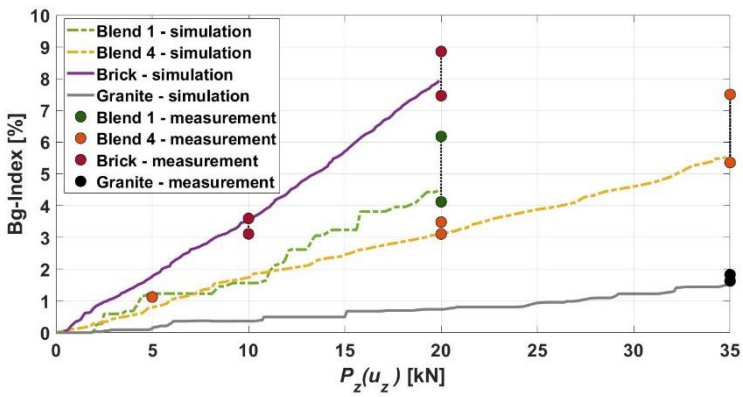


Figure 35. Bg-Index progress for Blend 1 and 4, Brick and Granite UGMs with respect to $P_z(u_z)$. (Paper IV)

3.6. Determination of Aggregate Fracture Evolution Parameter

In Paper V, an alternative aggregate fracture evolution parameter, capturing aggregate fracture within UGMs under increasing loading, was proposed based on combining confined uniaxial compression test results and DEM modelling, cf. also Subsection 2.2.5. The proposed fracture evolution parameter, $u_B(P_z)$, defined in Eq. (23) offers a potentially feasible method to capture the progression of aggregate fracture under applied loading in UGMs even for blended UGMs. Examples of how $u_B(P_z)$, is obtained are shown in Figure 36 by comparing hypothetical DEM modelled responses assuming no fracture with the mean measured curves for Granite36 (11.2 – 16) and Granite25 (2 – 16), with two P_{max} levels included for illustration. The response differs between the two UGMs. For Granite36, the hypothetical DEM response matches the mean measured curve only up to approximately 1.5 kN, while for Granite25 this agreement extends to about 9 kN, resulting in smaller differences at low load levels for the 2–16 mm UGM compared with the 11.2 – 16 mm UGM. This differences may be related to differences in aggregate fracture strength, gradation and the load level at which fracture initiates.

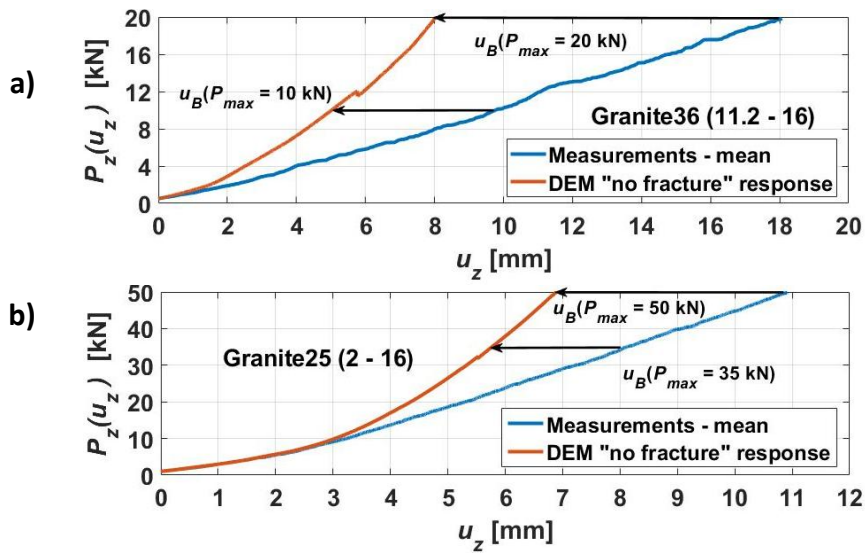


Figure 36. Hypothetical response without fracture alongside mean measurement curves for a) Granite36 (11.2 – 16), b) Granite25 (2 – 16). (Paper V)

In Figure 37, the relationship between Bg-Index and $u_B(P_{max})$ is examined for all UGMs and load levels. The measured Bg-Index at P_{max} is plotted as a function of $u_B(P_{max})$ in Figure 37(a), along with power-law fits for each gradation. An increase in Bg-Index with increasing $u_B(P_{max})$ is observed, although some scatter is present. The correlation between the individual gradations is strong, with coefficients of determination ($Corr$) exceeding 79% in all cases. This indicates that $u_B(P_{max})$ effectively captures the combined effects of aggregate strength, applied load magnitude, and their interaction.

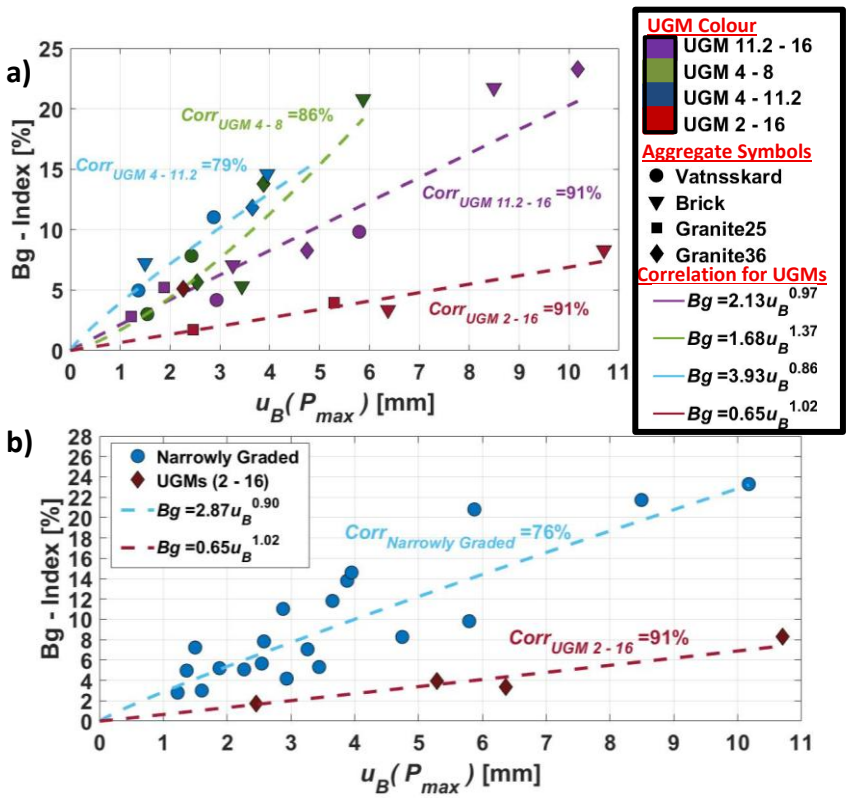


Figure 37. The relation between Bg-Index(P_{max}) and $u_B(P_{max})$ for a) all UGMs, and b) two regimes consisting of narrowly graded UGMs and UGMs with gradations of 2–16 mm. (Paper V)

In order to assess whether the effect of gradation can be partially accounted through $u_B(P_{max})$, Figure 37(b) presents the relationship between Bg-Index and $u_B(P_{max})$ for UGMs grouped in two categories: the densely

graded UGM (2–16 mm) and the UGMs with narrower gradations. Power-law fits for each group are shown using dashed lines and exhibit reasonably strong correlations, with $Corr$ exceeding 76%, allowing for prediction of Bg-Index from measured $u_B(P_{max})$ for UGMs with varying gradations. This grouping arises from limitations of the Bg-Index in capturing aggregate fracture in densely graded UGMs, where the extent of fracture may be underestimated. At a given Bg-Index level, the 2–16 mm UGMs display significantly higher $u_B(P_{max})$ values, up to five times of the narrowly graded UGMs.

The fracture evolution parameter $u_B(P_{max})$ also potentially offers a means to examine the development of aggregate fracture with increasing applied load without the need to interrupt testing for re-sieving of the UGMs. This is illustrated qualitatively in Figure 38 where $u_B(P_z)$ is presented for the Granite25 and Granite36 UGMs, together with their corresponding Bg-Indices determined for two P_{max} values. Both parameters are presented as average of three measurements and normalized by their respective maximum values for each UGM to enable direct comparison. As depicted in Figure 38, the normalized Bg-Index determined at first $P_{\max}$ and the corresponding $u_B(P_{max})$ at the same load level exhibit reasonable quantitative agreement. Clear differences in $u_B(P_z)$ evolution is also observed between each aggregate type: Granite36 UGMs show an increase in u_B almost from the onset of loading, whereas Granite25 UGMs exhibit a delayed response, with u_B increasing after reaching P_z = 8 kN. This pattern aligns qualitatively with the measured LA values of the two aggregate types. Furthermore, for Granite25 UGMs, the increase in $u_B(P_z)$ is slower for densely graded one, which is expected, as denser gradations would have more aggregate-to-aggregate contact points and thus more uniform load distribution within the UGMs. This aligns with the discussion of the results shown in Figure 14(c).

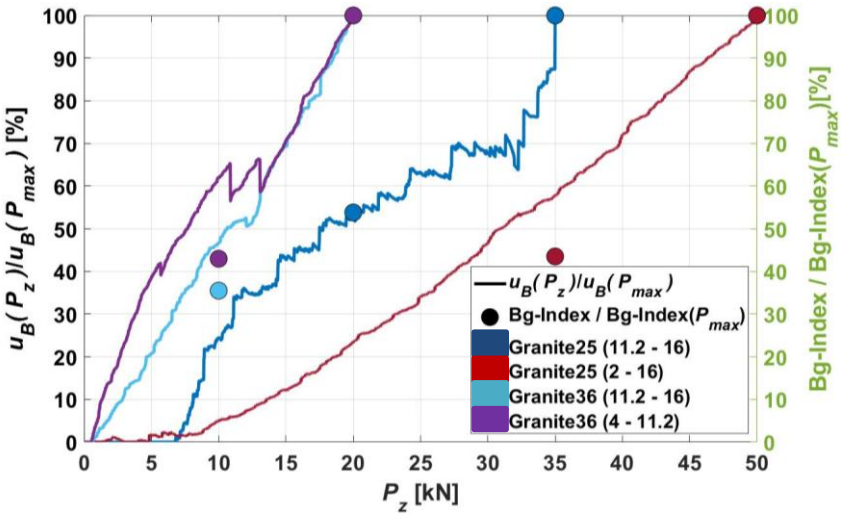


Figure 38. Normalized $u_B(P_z)$ and Bg-Index for Granite25 11.2 – 16, 2 – 16 and Granite36 11.2 – 16, 4 – 11.2. (Paper V)

The results shown in Figure 36 – 38 indicate that $u_B(P_z)$ captures reasonably well aggregate fracture in UGMs with varying gradations and under progressively applied loads. Consequently, $u_B(P_z)$ is a promising fracture evolution parameter, which may be used to identify allowable load levels for specific UGM gradations and aggregate types, thereby supporting optimization of road structural designs to avoid excessive aggregate fracture in unbound road layers.

4. Conclusions and Future Work

This thesis aims to improve the understanding, characterization, and quantitative prediction of aggregate fracture in unbound granular materials (UGMs) used in roads. A new DEM-based framework is developed and validated for UGMs with different gradations, blends, aggregate types—including marginal-quality aggregates—and loading magnitudes. The framework integrates novel particle contact laws and fracture models designed to accurately capture aggregate deformation and damage near contact points. This is essential for reliable extrapolation beyond the calibration range, particularly for marginal-quality materials such as crushed brick and volcanic aggregates, and for evaluating new UGMs and pavement designs.

For the UGMs composed of one type of granite aggregates, it is shown that combining the contact law developed by Olsson et al. (2019a; 2019b) with a particle fracture model based on the Weibull weakest-link approach captures the effects of gradation and axial compressive load on both the macromechanical response and aggregate fracture with reasonable accuracy. The simulated force–displacement response agrees with the measurements within 10% in most cases, with larger deviations, up to 20%, observed only for the 2–16 mm gradation after 40 kN of axial compression. The predicted extent of aggregate fracture is also in close quantitative and qualitative agreement with the experimental observations.

A new particle fracture model is developed to extend the DEM framework to a wider range of aggregate types. The model accounts for both aggregate shape variability and the statistical volume effect on the strength distribution of individual particles. Compared with two widely used fracture approaches—the classical Weibull distribution and the Weibull weakest-link formulation—it provides the best agreement with experimental observations for all aggregate types examined.

The model is then implemented in a DEM framework for confined uniaxial compression of UGMs. Its performance is evaluated for UGMs made of granite, volcanic rock, and crushed brick, across three gradations and under two compressive load levels. The simulations show strong agreement with the measured Bg-Index, with correlation coefficients above 90% in all cases.

The feasibility of using the developed DEM framework to optimize the placement of UGMs in unbound road layers, particularly those containing aggregates of marginal strength, is also investigated. This is done by simulating aggregate fracture in UGMs subjected to cyclic compressive loading representative of unbound layers in low-volume roads. It is shown that the model captures well both the influence of UGM placement on the extent of aggregate fracture and the influence of aggregate fracture on elastic and permanent deformation under monotonic and cyclic loading. The model thus shows potential for optimizing the structural design of unbound pavement layers and supporting the use of marginal-quality aggregates in road construction.

Aggregate fracture performance of blended UGMs containing granite as a conventional aggregate and brick as a marginal aggregate is evaluated experimentally in confined compression tests. The blends are found to be markedly stiffer than the pure brick UGM, and their Bg-Index consistently falls between those of the pure granite and pure brick UGMs. The aggregate fracture is also found to depend strongly on which size fraction is replaced, with substitution of the coarsest fraction producing about 70% higher Bg-Index than substitution of the finest fraction. The obtained experimental results are further used to assess the ability of the developed DEM framework to capture the effects of blended UGM composition and load magnitude on aggregate fracture. The framework is shown to capture the ranking of blends in terms of stiffness and fracture, as well as the load-dependent progression of crushing.

X-ray computed tomography (CT) is employed to further investigate aggregate breakage mechanisms and the spatial distribution of aggregate fracture in UGMs. In a blended UGM, it reveals clear differences in fracture patterns between the two aggregate types. Fractured granite is concentrated near the bottom of the specimen, whereas fractured brick is more uniformly distributed over the specimen height. The two aggregate types also differ in

their fragment-size distributions. Granite generally breaks into a few fragments retained on the next smaller sieve, while brick produces a broader fragment-size distribution, including fines.

Overall, the DEM framework developed in this thesis provides a useful tool for quantifying how load level, gradation, composition, and aggregate properties influence UGM performance, particularly aggregate crushing. The good agreement between experiments and simulations supports the proposed particle contact law and statistical fracture model as realistic representations of material behaviour. Beyond interpreting test results, the framework can be used to identify optimal aggregate blends and gradations, perform parametric studies, and determine critical load thresholds for fracture initiation and progression. It also provides a basis for evaluating new or marginal aggregates, assessing crushing susceptibility under different loading conditions, and supporting the design of UGMs for unbound pavement layers with reduced reliance on extensive laboratory testing.

Results from the confined monotonic compression tests are systematically analysed using statistical methods. It is shown that aggregate fracture in UGMs is governed by the combined effects of aggregate quality, gradation, and applied load. Aggregate quality and load are the dominant factors, with a significant interaction between them, while gradation also has a significant, though more moderate, effect. Fracture in UGMs can therefore not be reliably predicted from standard aggregate indices alone, but must be assessed under gradations and load levels representative of field conditions. To support such assessments, a new approach is introduced for characterizing aggregate fracture evolution in UGMs under progressively applied loading by combining experimental measurements with DEM modelling based on the experimental conditions, but excluding particle fracture. A new fracture evolution parameter, defined as the difference between measured and simulated UGM responses, is proposed to track fracture progression during compression and is found to correlate strongly with the experimental data for all aggregate types, gradations, and load levels examined.

Future work should focus on supporting implementation of the developed DEM framework for optimizing UGM material design and the structural design of unbound pavement layers. In particular, further research is needed to establish the link between DEM simulations and the long-term field

performance of UGMs, and to validate the framework under a wider range of conditions, loading scenarios, and UGM types. One particularly important application in this context is the development of guidelines for the use of marginal-quality aggregates in road construction, specifying allowable stress levels, aggregate contents, and the size fractions in which such materials can be incorporated.

Limitations in aggregate shape representation in the model should also be addressed, by exploring more complex geometries, such as clumps. This would improve the ability to capture the effects of aggregate shape on UGM mechanical behaviour and enable more detailed modelling of aggregate fragmentation. Additionally, UGM gradations encountered in field applications include a fraction of fine particles, which can become increasingly significant after aggregate fracture by affecting the interactions between aggregates within the unbound granular layer. Further research is needed to incorporate the fine particles into DEM modelling in computationally efficient manner. Similarly, under field conditions, aggregates are subjected to varying moisture levels, which influence the compressive behaviour of UGMs. Failing to account for these factors may limit predictions capability of DEM model. Finally, it must be pointed out that while single aggregate crushing tests provide a simple way for measuring distributions of aggregate fracture forces, they do not fully capture the behaviour of aggregates under contact loading, for instance with respect to the influence of contact points distribution on the aggregate surface. In the future, alternative tests methods, providing more thorough insight into the fracture characteristics of individual aggregates must be considered.

5. References

Afshar, T., Disfani, M. M., Arulrajah, A., Narsilio, G. A., & Emam, S. (2017). Impact of particle shape on breakage of recycled construction and demolition aggregates. *Powder Technology*, *308*, 1-12. https://doi.org/10.1016/j.powtec.2016.11.043

Agnolin, I., & Roux, J.-N. (2007). Internal states of model isotropic granular packings. III. Elastic properties. *Physical Review E*, *76*(6), 061304. https://doi.org/10.1103/PhysRevE.76.061304

Arulrajah, A., Piratheepan, J., Disfani, M. M., & Bo, M. W. (2013). Geotechnical and Geoenvironmental Properties of Recycled Construction and Demolition Materials in Pavement Subbase Applications. *Journal of Materials in Civil Engineering*, *25*(8), 1077-1088. https://doi.org/10.1061/(ASCE)MT.1943-5533.0000652

ASTM C127. (2015). *Standard Test Method for Relative Density (Specific Gravity) and Absorption of Coarse Aggregate* ASTM International, West Conshohocken, PA. https://doi.org/10.1520/C0127-12

Barbhuiya, S., Qureshi, T., & Das, B. B. (2025). Advancing sustainable pavements: a review of low-carbon construction materials and practices. *Discover Concrete and Cement*, *1*(1), 19. https://doi.org/10.1007/s44416-025-00020-w

Benard, A., & Bos-Levenbach, E. (1955). *The plotting of observations on probability-paper*. Stichting Mathematisch Centrum. Statistische Afdeling.

Bisht, M. S., & Das, A. (2021). DEM Study on Particle Shape Evolution during Crushing of Granular Materials. *International Journal of Geomechanics*, *21*(7), 04021101. https://doi.org/doi:10.1061/(ASCE)GM.1943-5622.0002067

Bjarnason, G., Erlingsson, S., & Pétursson, P. (1999). Aggregate resistance to fragmentation. weathering and abrasion. *BUSL–co-operation in the field of road research in Iceland E-37 p*, *34*.

Cervera, C. C., Jelagin, D., Partl, M. N., & Larsson, P.-L. (2017). Contact-induced deformation and damage of rocks used in pavement materials. *Materials & Design*, *133*, 255-265. https://doi.org/10.1016/j.matdes.2017.08.003

Cheng, Y. P., Nakata, Y., & Bolton, M. D. (2003). Discrete element simulation of crushable soil. *Geotechnique*, *53*(7), 633-641. https://doi.org/10.1680/geot.2003.53.7.633

Colagrande, S., Ranalli, D., & Tallini, M. (2011). Ground Penetrating Radar Assessment of Flexible Road Pavement Degradation. *International Journal of Geophysics*, *2011*(1), 989136. https://doi.org/10.1155/2011/989136

Cole, D. M., & Peters, J. F. (2008). Grain-scale mechanics of geologic materials and lunar simulants under normal loading. *Granular Matter*, *10*(3), 171-185. https://doi.org/10.1007/s10035-007-0066-y

Cui, L., & Bhattacharya, S. (2017). Choice of aggregates for permeable pavements based on laboratory tests and DEM simulations. *International Journal of Pavement Engineering*, *18*(2), 160-168. https://doi.org/10.1080/10298436.2015.1053480

Cundall, P. A., & Strack, O. D. L. (1979). A discrete numerical model for granular assemblies. *Geotechnique*, *29*(1), 47-65. https://doi.org/10.1680/geot.1979.29.1.47

de Andrade Salgado, F., & de Andrade Silva, F. (2022). Recycled aggregates from construction and demolition waste towards an application on structural concrete: A review. *Journal of Building Engineering*, *52*, 104452. https://doi.org/10.1016/j.jobe.2022.104452

de Bono, J., & McDowell, G. (2016). Particle breakage criteria in discrete-element modelling. *Geotechnique*, *66*(12), 1014-1027. https://doi.org/10.1680/jgeot.15.P.280

Diagne, M., Tinjum, J. M., & Nokkaew, K. (2015). The effects of recycled clay brick content on the engineering properties, weathering durability, and resilient modulus of recycled concrete aggregate. *Transportation Geotechnics*, *3*, 15-23. https://doi.org/10.1016/j.trgeo.2014.12.003

Ellis, S., Goodwin, A., Laycock, E., & Hurst, C. (2015). Use of crushed brick in reinforced earth railway structures. *Proceedings of the Institution of Civil Engineers - Construction Materials*, *169*(2), 93-105. https://doi.org/10.1680/coma.15.00014

Field, A. (2024). *Discovering statistics using IBM SPSS statistics*. Sage publications limited.

Guo, Y., Zhao, C., Markine, V., Jing, G., & Zhai, W. (2020). Calibration for discrete element modelling of railway ballast: A review. *Transportation Geotechnics*, *23*, 100341. https://doi.org/https://doi.org/10.1016/j.trgeo.2020.100341

Hertz, H. (1881). Über die Berührung fester elastischer Körper. *J reine und angewandte Mathematik*, *92*, 156.

Itasca Consulting Group. (2019). *PFC3D (particle flow code in 3 dimensions), Version 6.0*. Itasca Consulting Group, Inc.

Kong, K.-f., Chen, F., Xiao, Y.-j., Li, Y.-b., Jiang, Y., Wang, M., Tutumluer, E., Chen, Y.-l., & Li, Z.-y. (2023). Investigating particle breakage and compressibility characteristics of unbound aggregate materials recycled from building demolition wastes. *Journal of Central South University*, *30*(10), 3499-3516. https://doi.org/10.1007/s11771-023-5463-9

Lu, M., & McDowell, G. R. (2010). Discrete element modelling of railway ballast under monotonic and cyclic triaxial loading. *Geotechnique*, *60*(6), 459-467. https://doi.org/10.1680/geot.2010.60.6.459

Mahmoud, E., & Masad, E. (2007). Experimental Methods for the Evaluation of Aggregate Resistance to Polishing, Abrasion, and Breakage. *Journal of Materials in Civil Engineering*, *19*(11), 977-985. https://doi.org/10.1061/(ASCE)0899-1561(2007)19:11(977)

Marsal, R. J., Moreno Gomez, E., Nuñez, G. A., Cuellar, B. R., & Moreno, R. R. (1965). *Research on the Behaviour of Granular Soils and Rockfill Samples* Mexico.

Mindlin, R. D., & Deresiewicz, H. (1953). Elastic Spheres in Contact Under Varying Oblique Forces. *Journal of applied mechanics*, *20*(3), 327-344. https://doi.org/10.1115/1.4010702

Moretti, L., Di Mascio, P., & Panunzi, F. (2012). Economic Sustainability of Concrete Pavements. *Procedia - Social and Behavioral Sciences*, *53*, 125-133. https://doi.org/10.1016/j.sbspro.2012.09.866

Olsson, E., Jelagin, D., & Forquin, P. A. (2019a). Computational framework for analysis of contact-induced damage in brittle rocks. *International Journal of Solids and Structures*, *167*, 24-35. https://doi.org/10.1016/j.ijsolstr.2019.03.001

Olsson, E., Jelagin, D., & Partl, M. N. (2019b). New discrete element framework for modelling asphalt compaction. *Road Materials and Pavement Design*, *20*(sup2), S604-S616. https://doi.org/10.1080/14680629.2019.1633750

Olsson, E., Jelagin, D., & Partl, M. N. (2020). Numerical Evaluation of Crushing Resistance of Unbound Road Material. In C. Raab, *Proceedings of the 9th International Conference on Maintenance and Rehabilitation of Pavements—Mairepav9* Cham.https://doi.org/10.1007/978-3-030-48679-2_20

Olsson, E., & Larsson, P. L. (2015). Micromechanical investigation of the fracture behavior of powder materials. *Powder Technology*, *286*, 288-302. https://doi.org/10.1016/j.powtec.2015.08.018

Oskooei, P. R., Mohammadinia, A., Arulrajah, A., & Horpibulsuk, S. (2021b). DEM modeling and experimental analysis of the breakage behavior of recycled crushed brick particles. *Transportation Geotechnics*, *30*, 100586. https://doi.org/10.1016/j.trgeo.2021.100586

Oskooei, P. R., Mohammadinia, A., Arulrajah, A., Horpibulsuk, S., & Emam, S. (2021a). Crushing behavior of recycled waste materials: Experimental analysis and DEM simulation. *Construction and Building Materials*, *299*, 124226. https://doi.org/10.1016/j.conbuildmat.2021.124226

Partl, M. N. (2019). Quest for improving service life of asphalt roads. *RILEM Technical Letters*, *4*, 154-162. https://doi.org/https://doi.org/10.21809/rilemtechlett.2019.102

Ragnoli, A., De Blasiis, M. R., & Di Benedetto, A. (2018). Pavement Distress Detection Methods: A Review. *Infrastructures*, *3*(4), 58. https://www.mdpi.com/2412-3811/3/4/58

Saeed, A., Hall Jr, J., & Barker, W. (2001). *Performance-related tests of aggregates for use in unbound pavement layers*. National Academy Press.

Steypustöðin ehf. (2023). *Environmental Product Declaration: Vatnsskard Quarry* Steypustöðin, https://steypustodin.is/wp-content/uploads/2023/06/EPD-Vatnsskard-Steypustodin-Namur.pdf

Sveinsdottir, E., & Wigum, B. (2003). *IAEG C-17 Aggregates* Technical report,

Vallejo, L. E., Lobo-Guerrero, S., & Hammer, K. (2006). Degradation of a Granular Base under a Flexible Pavement: DEM Simulation. *International Journal of Geomechanics*, *6*(6), 435-439. https://doi.org/10.106(ASCE)1532-3641(2006)6:6(435)

Weibull, W. (1951). A statistical distribution function of wide applicability. *Journal of applied mechanics*.

Wu, Y., Parker, F., & Kandhal, P. S. (1998). Aggregate Toughness/Abrasion Resistance and Durability/Soundness Tests Related to Asphalt Concrete Performance in Pavements. *Transportation Research Record*, *1638*(1), 85-93. https://doi.org/10.3141/1638-10

Zeghal, M. (2009). The Impact of Grain Crushing on Road Performance. *Geotechnical and Geological Engineering*, *27*(4), 549-558. https://doi.org/10.1007/s10706-009-9256-1

Zhang, H.-L., Tang, Y., Meng, T., & Zhan, L.-T. (2021). Evaluating the crushing characteristics of recycled construction and demolition waste for use in road bases. *Transportation Geotechnics*, *28*, 100543. https://doi.org/10.1016/j.trgeo.2021.100543